



Universidade de Brasília

# Brasília Military Middle School 2017

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1 Contest

2 Theory

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7 Strings

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Contest (1)

template.cpp 28 lines

```
#include <bits/stdc++.h>
using namespace std;
// Ours
#define endl '\n'
#define debug(x...) cout<<"["#x"]: ", [] (auto...$) {{(cout<<$<<"<
    "),...);} (x),cout<<endl
// KACTL
typedef long long ll;
typedef long double ld;
#define sz(x) (int)x.size()
#define all(x) x.begin(), x.end()
#define vi vector<int>
#define pii pair<int, int>
#define rep(i, a, b) for(int i = (a); i < (b); i++)
#define pb push_back
#define eb emplace_back

constexpr int oo = (((unsigned int)-1)>>2);

void solve(){
    int n; cin >> n;
}

int main() {
    cin.tie(0)->sync_with_stdio(0);
    int t = 1;
    // cin >> t;
    while (t--) solve();
}
```

.bashrc 21 lines

```
teste() {
    for f in "$2"/*; do
        echo "=== $f"
        cat $f
        echo "=== Out"
        $1 < $f
    done
}
```

```
1 # teste ./b bf
stress() {
1     for ((i=1; ; i++)) do
        echo "Test $i"
        echo $i | $1 > gen_in
2     $2 < gen_in > main_out
        $3 < gen_in > brute_out
        if ! diff -w main_out brute_out > /dev/null; then
6         return 1
        fi
    done
11 }
# stress "python3 gen.py" ./main ./brute
```

.vimrc 5 lines

```
22 set autoindent smartindent ts=4 sw=4 rnu nu
    " Select region and then type :Hash to hash your selection.
23 " Useful for verifying that there aren't mistypes.
    ca Hash w !cpp -dD -P -fpreprocessed \ | tr -d '[:space:]' \
        \ | md5sum \ | cut -c-6
```

hash.sh 3 lines

```
# Hashes a file, ignoring all whitespace and comments. Use for
# verifying that code was correctly typed.
cpp -dD -P -fpreprocessed | tr -d '[:space:]' | md5sum |cut -c-6
```

Makefile 1 lines

```
CXXFLAGS := -std=c++20 -Wall -Wextra -Wshadow -Wconversion -
fsanitize=address,undefined -g3
```

Theory (2)

2.1 Geometry

2.1.1 Trigonometry

(V + W) tan(v − w)/2 = (V − W) tan(v + w)/2

where V, W are lengths of sides opposite angles v, w.

a cos x + b sin x = r cos(x − ϕ)

a sin x + b cos x = r sin(x + ϕ)

where  $r = \sqrt{a^2 + b^2}$ ,  $\phi = \text{atan2}(b, a)$ .

2.1.2 Triangles

Length of median (divides triangle into two equal-area triangles):

$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Length of bisector (divides angles in two):

$s_a = \sqrt{bc \left[ 1 - \left( \frac{a}{b+c} \right)^2 \right]}$

Law of tangents:  $\frac{a+b}{a-b} = \frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}}$

2.1.3 Quadrilaterals

With side lengths  $a, b, c, d$ , diagonals  $e, f$ , diagonals angle  $\theta$ , area  $A$  and magic flux  $F = b^2 + d^2 - a^2 - c^2$ :

$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2 f^2 - F^2}$

For cyclic quadrilaterals the sum of opposite angles is 180°,  $ef = ac + bd$ , and  $A = \sqrt{(p-a)(p-b)(p-c)(p-d)}$  ( $p$  is semiperimeter).

2.2 Combinatorics

2.2.1 Permutations

Cycles

Let  $g_S(n)$  be the number of  $n$ -permutations whose cycle lengths all belong to the set  $S$ . Then

$$\sum_{n=0}^{\infty} g_S(n) \frac{x^n}{n!} = \exp \left( \sum_{n \in S} \frac{x^n}{n} \right)$$

Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1) + D(n-2)) = nD(n-1) + (-1)^n = \left\lfloor \frac{n!}{e} \right\rfloor$$

2.2.2 Partitions and subsets

Partition function

Number of ways of writing  $n$  as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k-1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

|        |   |   |   |   |   |   |    |    |    |    |     |      |      |
|--------|---|---|---|---|---|---|----|----|----|----|-----|------|------|
| $n$    | 0 | 1 | 2 | 3 | 4 | 5 | 6  | 7  | 8  | 9  | 20  | 50   | 100  |
| $p(n)$ | 1 | 1 | 2 | 3 | 5 | 7 | 11 | 15 | 22 | 30 | 627 | ~2e5 | ~2e8 |

Lucas' Theorem

Let  $n, m$  be non-negative integers and  $p$  a prime. Write  $n = n_k p^k + \dots + n_1 p + n_0$  and  $m = m_k p^k + \dots + m_1 p + m_0$ . Then  $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$ .

2.2.3 General purpose numbers

Bernoulli numbers

EGF of Bernoulli numbers is  $B(t) = \frac{t}{e^t - 1}$  (FFT-able).  $B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$

Sums of powers:

$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^{\infty} f(i) = \int_m^{\infty} f(x)dx - \sum_{k=1}^{\infty} \frac{B_k}{k!} f^{(k-1)}(m)$$
$$\approx \int_m^{\infty} f(x)dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

### Stirling numbers of the first kind

Number of permutations on  $n$  items with  $k$  cycles.

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k), \; c(0,0) = 1$$
$$\sum_{k=0}^n c(n,k)x^k = x(x+1)\dots(x+n-1)$$

$$c(8,k) = 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$$
$$c(n,2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$$

### Eulerian numbers

Number of permutations of  $n$  elements with exactly  $k$  descents (equivalently,  $k$  ascents or  $k$  excedances).

$$E(n,k) = (n-k)E(n-1,k-1) + (k+1)E(n-1,k)$$
$$E(n,0) = E(n,n-1) = 1$$
$$E(n,k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

### Stirling numbers of the second kind

Partitions of  $n$  distinct elements into exactly  $k$  groups.

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$
$$S(n,1) = S(n,n) = 1$$
$$S(n,k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

### Bell numbers

Total number of partitions of  $n$  distinct elements.  $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$  For  $p$  prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod{p}$$

## 2.3 Number Theory

### 2.3.1 Maximum number of divisors

Maximum  $\tau(n)$  for  $n \leq N$ ; last column gives one such  $n$ .

| $N$    | $\tau$ | $n$       | $N$       | $\tau$ | $n$                |
|--------|--------|-----------|-----------|--------|--------------------|
| $10^2$ | 12     | 60        | $10^{10}$ | 2304   | 6983776800         |
| $10^3$ | 32     | 840       | $10^{11}$ | 4032   | 97772875200        |
| $10^4$ | 64     | 7560      | $10^{12}$ | 6720   | 963761198400       |
| $10^5$ | 128    | 83160     | $10^{13}$ | 10752  | 9316358251200      |
| $10^6$ | 240    | 720720    | $10^{14}$ | 17280  | 97821761637600     |
| $10^7$ | 448    | 8648640   | $10^{15}$ | 26880  | 866421317361600    |
| $10^8$ | 768    | 73513440  | $10^{16}$ | 41472  | 8086598962041600   |
| $10^9$ | 1344   | 735134400 | $10^{17}$ | 64512  | 74801040398884800  |
|        |        |           | $10^{18}$ | 103680 | 897612484786617600 |

## 2.4 Graphs

### 2.4.1 Erdős–Gallai theorem

A simple graph with node degrees  $d_1 \geq \dots \geq d_n$  exists iff  $d_1 + \dots + d_n$  is even and for every  $k = 1 \dots n$ ,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i,k).$$

### 2.4.2 Maximum Independent Set

Just find a max clique of the complement. If the graph is bipartite, see MinimumVertexCover.h.

### 2.4.3 Min Cut

After running a max flow, the left side of a min-cut from  $s$  to  $t$  is given by all vertices reachable from  $s$ , only traversing edges with positive residual capacity.

## Math (3)

## 3.1 Number Theory

### 3.1.1 Modular Arithmetic

| ModPow.h                                                                                                                                                                                                        | b83e45, 7L, 1m10s |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <pre>const ll mod = 1000000007; // faster if const ll modpow(ll b, ll e) {     ll ans = 1;     for (; e; b = b * b % mod, e /= 2)         if (e &amp; 1) ans = ans * b % mod;     return ans; } // d1ec54</pre> |                   |

| ModMulLL.h                                                                                                              |  |
|-------------------------------------------------------------------------------------------------------------------------|--|
| <b>Description:</b> Calculate $a \cdot b \bmod c$ (or $a^b \bmod c$ ) for $0 \leq a, b \leq c \leq 7.2 \cdot 10^{18}$ . |  |
| <b>Time:</b> $\mathcal{O}(1)$ for modmul, $\mathcal{O}(\log b)$ for modpow                                              |  |
| bbbdb8f, 11L, 2m                                                                                                        |  |

| typedef unsigned long long ull;                                                                                                                                                                                                                                                                                                                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre>ull modmul(ull a, ull b, ull M) {     ll ret = a * b - M * ull(1.L / M * a * b);     return ret + M * (ret &lt; 0) - M * (ret &gt;= (1L)M); } // e9309c ull modpow(ull b, ull e, ull mod) {     ull ans = 1;     for (; e; b = modmul(b, b, mod), e /= 2)         if (e &amp; 1) ans = modmul(ans, b, mod);     return ans; } // 100b91</pre> |  |

| ModLog.h                                                                                                                                                                        |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Description:</b> Returns the smallest $x > 0$ s.t. $a^x = b \pmod m$ , or $-1$ if no such $x$ exists. <code>modLog(a,1,m)</code> can be used to calculate the order of $a$ . |  |
| <b>Memory:</b> $\mathcal{O}(\sqrt{m})$                                                                                                                                          |  |
| <b>Time:</b> $\mathcal{O}(\sqrt{m})$                                                                                                                                            |  |

c040b8, 11L, 2m10s

|                                                                                                                                                                                                                                                                                           |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre>ll modLog(ll a, ll b, ll m) {     ll n = (1L) sqrt(m) + 1, e = 1, f = 1, j = 1;     unordered_map&lt;ll, ll&gt; A;     while (j &lt;= n &amp;&amp; (e = f = e * a % m) != b % m)         A[e * b % m] = j++;     if (e == b % m) return j;     if (__gcd(m, e) == __gcd(m, b))</pre> |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|                                                                                                     |  |
|-----------------------------------------------------------------------------------------------------|--|
| <pre>rep(i,2,n+2) if (A.count(e = e * f % m))     return n * i - A[e]; return -1; } // c040b8</pre> |  |
|-----------------------------------------------------------------------------------------------------|--|

# ModSqrt.h

**Description:** Tonelli-Shanks algorithm for modular square roots. Finds  $x$  s.t.  $x^2 = a \pmod p$  ( $-x$  gives the other solution).

**Time:**  $\mathcal{O}(\log^2 p)$  worst case,  $\mathcal{O}(\log p)$  for most  $p$

---

"ModPow.h"

45c733, 22L, 4m30s

```

const ll p = mod; // mod is p
ll modSqrt(ll a) {
    if(p==2) return a&1;
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    assert(modpow(a, (p-1)/2) == 1); // else no solution
    if (p % 4 == 3) return modpow(a, (p+1)/4);
    // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
    ll s = p - 1, n = 2;
    int r = 0, m;
    while (s % 2 == 0) ++r, s /= 2;
    while (modpow(n, (p - 1) / 2) != p - 1) ++n;
    ll x = modpow(a, (s + 1) / 2);
    ll b = modpow(a, s), g = modpow(n, s);
    for (; r = m) {
        ll t = b;
        for (m = 0; m < r && t != 1; ++m) t = t * t % p;
        if (m == 0) return x;
        ll gs = modpow(g, 1LL << (r - m - 1));
        g = gs * gs % p; x = x * gs % p; b = b * g % p;
    } // d569f9
} // d2ac0b

```

### CRT.h

**Description:** Chinese Remainder Theorem.

crt(a, m, b, n) computes  $x$  such that  $x \equiv a \pmod{m}$ ,  $x \equiv b \pmod{n}$ . If  $|a| < m$  and  $|b| < n$ ,  $x$  will obey  $0 \leq x < \text{lcm}(m, n)$ . Assumes  $mn < 2^{62}$ .

**Time:**  $\log(n)$

| "euclid.h"                                                                                                                                                                                                                                                             | 04d93a, 7L, 1m30s |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <pre>ll crt(ll a, ll m, ll b, ll n) {     if (n &gt; m) swap(a, b), swap(m, n);     ll x, y, g = euclid(m, n, x, y);     assert((a - b) % g == 0); // else no solution     x = (b - a) % n * x % n / g * m + a;     return x &lt; 0 ? x + m*n/g : x; } // 04d93a</pre> |                   |

### 3.1.2 Primality and Divisibility

| FastEratosthenes.h                                                          |                 |
|-----------------------------------------------------------------------------|-----------------|
| <b>Description:</b> Prime sieve for generating all primes smaller than LIM. |                 |
| <b>Time:</b> LIM=1e9 $\approx$ 1.5s                                         | 6b2912, 20L, 4m |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 6b2912, 20L, 4m |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <pre>const int LIM = 1e6; bitset&lt;LIM&gt; isPrime; vi eratosthenes() {     const int S = (int)round(sqrt(LIM)), R = LIM / 2;     vi pr = {2}, sieve(S+1); pr.reserve((int)(LIM/log(LIM)*1.1));     vector&lt;pii&gt; cp;     for (int i = 3; i &lt;= S; i += 2) if (!sieve[i]) {         cp.push_back({i, i * i / 2});         for (int j = i * i; j &lt;= S; j += 2 * i) sieve[j] = 1;     } // d22e52     for (int L = 1; L &lt;= R; L += S) {         array&lt;bool, S&gt; block{};         for (auto &amp;[p, idx] : cp)             for (int i=idx; i &lt; S+L; idx = (i+=p)) block[i-L] = 1;         rep(i,0,min(S, R - L))</pre> |                 |

```
        if (!block[i]) pr.push_back((L + i) * 2 + 1);
    } // 5b6623
    for (int i : pr) isPrime[i] = 1;
    return pr;
} // 8ee6d2
```

### MillerRabin.h

**Description:** Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to  $7 \cdot 10^{18}$ ; for larger numbers, use Python and extend A randomly.

**Time:** 7 times the complexity of  $a^b \bmod c$ .

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "ModMulL.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 60dcd1, 12L, 2m30s |
| <pre>bool isPrime(ull n) {     if (n &lt; 2    n % 6 % 4 != 1) return (n   1) == 3;     ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022},         s = __builtin_ctzll(n-1), d = n &gt;&gt; s;     for (ull a : A) { // ^count trailing zeroes         ull p = modpow(a%n, d, n), i = s;         while (p != 1 &amp;&amp; p != n-1 &amp;&amp; a % n &amp;&amp; i--)             p = modmul(p, p, n);         if (p != n-1 &amp;&amp; i != s) return 0;     } // edfaf1     return 1; } // 60dcd1</pre> |                    |

### PollardRho.h

**Description:** Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

**Time:**  $\mathcal{O}\left(n^{1/4}\right)$ , less for numbers with small factors.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "ModMulL.h", "MillerRabin.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | d8d98d, 18L, 3m20s |
| <pre>ull pollard(ull n) {     ull x = 0, y = 0, t = 30, prd = 2, i = 1, q;     auto f = [&amp;](ull x) { return modmul(x, x, n) + i; };     while (t++ % 40    __gcd(prd, n) == 1) {         if (x == y) x = ++i, y = f(x);         if ((q = modmul(prd, max(x,y) - min(x,y), n))) prd = q;         x = f(x), y = f(f(y));     } // 989d40     return __gcd(prd, n); } // cd2ac3 vector&lt;ull&gt; factor(ull n) {     if (n == 1) return {};     if (isPrime(n)) return {n};     ull x = pollard(n);     auto l = factor(x), r = factor(n / x);     l.insert(l.end(), all(r));     return l; } // d54ba8</pre> |                    |

### euclid.h

**Description:** Finds two integers  $x$  and  $y$ , such that  $ax + by = \gcd(a, b)$ . If you just need gcd, use the built in `__gcd` instead. If  $a$  and  $b$  are coprime, then  $x$  is the inverse of  $a \pmod b$ .

```
ll euclid(ll a, ll b, ll &x, ll &y) {
    if (!b) return x = 1, y = 0, a;
    ll d = euclid(b, a % b, y, x);
    return y -= a/b * x, d;
} // 33ba8f
```

## 3.2 Discrete Math

### 3.2.1 Convolutions

### FastFourierTransform.h

**Description:**  $\text{fft}(a)$  computes  $f(k) = \sum_x a[x] \exp(2\pi i \cdot kx/N)$  for all  $k$ .  $N$  must be a power of 2. Useful for convolution:  $\text{conv}(a, b) = c$ , where  $c[x] = \sum a[i]b[x-i]$ . For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by  $n$ , reverse(start+1, end), FFT back. Rounding is safe if  $(\sum a_i^2 + \sum b_i^2) \log_2 N < 9 \cdot 10^{14}$  (in practice  $10^{16}$ ; higher for random inputs). Otherwise, use NTT/FFTMod.

**Time:**  $\mathcal{O}(N \log N)$  with  $N = |A| + |B|$  ( $\sim 1s$  for  $N = 2^{22}$ )

```
typedef complex<double> C;
typedef vector<double> vd;
void fft(vector<C>& a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1); // (10% faster if double)
    static vector<C> rt(2, 1);
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n); rt.resize(n);
        auto x = polar(1.0L, acos(-1.0L) / k);
        rep(i, k, 2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
    } // a8a74e
    vi rev(n);
    rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j, 0, k) {
            auto x = (double *)&rt[j+k], y = (double *)&a[i+j+k];
            C z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
            a[i + j + k] = a[i + j] - z;
            a[i + j] += z;
        } // 69f11b
    } // 2258f7
}

vd conv(const vd& a, const vd& b) {
    if (a.empty() || b.empty()) return {};
    vd res(sz(a) + sz(b) - 1);
    int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
    vector<C> in(n), out(n);
    copy(all(a), begin(in));
    rep(i, 0, sz(b)) in[i].imag(b[i]);
    fft(in);
    for (C& x : in) x *= x;
    rep(i, 0, n) out[i] = in[-i & (n - 1)] - conj(in[i]);
    fft(out);
    rep(i, 0, sz(res)) res[i] = imag(out[i]) / (4 * n);
    return res;
} // 873509
```

### FastFourierTransformMod.h

**Description:** Higher precision FFT, can be used for convolutions modulo arbitrary integers as long as  $N \log_2 N \cdot \text{mod} < 8.6 \cdot 10^{14}$  (in practice  $10^{16}$  or higher). Inputs must be in  $[0, \text{mod})$ .

**Time:**  $\mathcal{O}(N \log N)$ , where  $N = |A| + |B|$  (twice as slow as NTT or FFT)

```
FastFourierTransform.h" b82773, 22L, 5m
typedef vector<ll> vl;
template<int M> vl convMod(const vl &a, const vl &b) {
    if (a.empty() || b.empty()) return {};
    vl res(sz(a) + sz(b) - 1);
    int B=32-__builtin_clz(sz(res)), n=1<<B, cut=int(sqrt(M));
    vector<C> L(n), R(n), outs(n), outl(n);
    rep(i,0,sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
    rep(i,0,sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
    fft(L), fft(R);
    rep(i,0,n) {
        int j = -i & (n - 1);
        outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
    } // cb3017
    fft(outl), fft(outs);
    rep(i,0,sz(res)) {
```

```
        ll av = ll(real(outl[i])+.5), cv = ll(imag(outs[i])+.5);
        ll bv = ll(imag(outl[i])+.5) + ll(real(outs[i])+.5);
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    } // 58fa4f
    return res;
} // c1f2f1
```

### NumberTheoreticTransform.h

**Description:**  $\text{ntt}(a)$  computes  $f(k) = \sum_x a[x]g^{xk}$  for all  $k$ , where  $g = \text{root}^{(mod-1)/N}$ .  $N$  must be a power of 2. Useful for convolution modulo specific nice primes of the form  $2^a b + 1$ , where the convolution result has size at most  $2^a$ . For arbitrary modulo, see FFTMod.  $\text{conv}(a, b) = c$ , where  $c[x] = \sum a[i]b[x-i]$ . For manual convolution: NTT the inputs, multiply pointwise, divide by  $n$ , reverse(start+1, end), NTT back. Inputs must be in  $[0, \text{mod})$ .

**Time:**  $\mathcal{O}(N \log N)$

```

a46443, 35L, 7m50s
const ll root = 62; // change mod to (119 << 23) + 1 =
998244353
// For p < 2^30 there is also e.g. 5 << 25, 7 << 26, 479 << 21
// and 483 << 21 (same root). The last two are > 10^9.
typedef vector<ll> vl;
void ntt(vl &a) {
    int n = sz(a), L = 31 - __builtin_clz(n);
    static vl rt(2, 1);
    for (static int k = 2, s = 2; k < n; k *= 2, s++) {
        rt.resize(n);
        ll z[] = {1, modpow(root, mod >> s)};
        rep(i,k,2*k) rt[i] = rt[i / 2] * z[i & 1] % mod;
    } // f39127
    vi rev(n);
    rep(i,0,n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
    rep(i,0,n) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k) rep(j,0,k) {
            ll z = rt[j + k] * a[i + j + k] % mod, &ai = a[i + j];
            a[i + j + k] = ai - z + (z > ai ? mod : 0);
            ai += (ai + z >= mod ? z - mod : z);
        } // 9a8565
    } // 3b763b
}
vl conv(const vl &a, const vl &b) {
    if (a.empty() || b.empty()) return {};
    int s = sz(a) + sz(b) - 1, B = 32 - __builtin_clz(s),
        n = 1 << B;
    int inv = modpow(n, mod - 2);
    vl L(a), R(b), out(n);
    L.resize(n), R.resize(n);
    ntt(L), ntt(R);
    rep(i,0,n)
        out[-i & (n - 1)] = (ll)L[i] * R[i] % mod * inv % mod;
    ntt(out);
    return {out.begin(), out.begin() + s};
} // 3876bf

```

### FastSubsetTransform.h

**Description:** Transform to a basis with fast convolutions of the form  $c[z] = \sum_{z=x \oplus y} a[x] \cdot b[y]$ , where  $\oplus$  is one of AND, OR, XOR. The size of  $a$  must be a power of two, and  $a$  and  $b$  must have the same power-of-two size.

**Time:**  $\mathcal{O}(N \log N)$

```
void FST(vi& a, bool inv) {
    for (int n = sz(a), step = 1; step < n; step *= 2) {
        for (int i = 0; i < n; i += 2 * step) rep(j,i,i+step) {
            int &u = a[j], &v = a[j + step]; tie(u, v) =
                inv ? pii(v - u, u) : pii(v, u + v); // AND
            inv ? pii(v, u - v) : pii(u + v, u); // OR
        }
    }
}
```

```
        pii(u + v, u - v); // XOR
    } // 7b7e71
} // faec61
if (inv) for (int& x : a) x /= sz(a); // XOR only
} // 57eeaf
vi conv(vi a, vi b) {
    FST(a, 0); FST(b, 0);
    rep(i,0,sz(a)) a[i] *= b[i];
    FST(a, 1); return a;
} // 3cbd18
```

3.2.2 Recurrences and Sums

**BerlekampMassey.h**  
**Description:** Recovers any  $n$ -order linear recurrence relation from the first  $2n$  terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. The idea should generalize to any field, but numerical stability for floats is not guaranteed. Output will have size  $\leq n$ .  
**Usage:** berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}  
**Time:**  $\mathcal{O}(N^2)$

---

"/math/ModPow.h" 96548b, 17L, 3m20s

```
vector<ll> berlekampMassey(vector<ll> s) {
    int n = sz(s), L = 0, m = 0;
    vector<ll> C(n), B(n), T;
    C[0] = B[0] = 1; ll b = 1;
    rep(i,0,n) { ++m;
        ll d = s[i] % mod;
        rep(j,1,L+1) d = (d + C[j] * s[i - j]) % mod;
        if (!d) continue;
        T = C; ll coef = d * modpow(b, mod-2) % mod;
        rep(j,m,n) C[j] = (C[j] - coef * B[j - m]) % mod;
        if (2 * L > i) continue;
        L = i + 1 - L; B = T; b = d; m = 0;
    } // 8c2376
    C.resize(L + 1); C.erase(C.begin());
    for (ll& x : C) x = (mod - x) % mod;
    return C;
} // 96548b
```

**LinearRecurrence.h**  
**Description:** Generates the  $k$ 'th term of an  $n$ -order linear recurrence  $S[i] = \sum_j S[i - j - 1]tr[j]$ , given  $S[0 \dots \geq n - 1]$  and  $tr[0 \dots n - 1]$ . Faster than matrix multiplication. Useful together with Berlekamp–Massey.  
**Usage:** linearRec({0, 1}, {1, 1}, k) //  $k$ 'th Fibonacci number  
**Time:**  $\mathcal{O}(n^2 \log k)$

---

f4e444, 21L, 3m50s

```
typedef vector<ll> Poly;
ll linearRec(Poly S, Poly tr, ll k) {
    int n = sz(tr);
    auto combine = [&](Poly a, Poly b) {
        Poly res(n * 2 + 1);
        rep(i,0,n+1) rep(j,0,n+1)
            res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
        for (int i = 2 * n; i > n; --i) rep(j,0,n)
            res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
        res.resize(n + 1);
        return res;
    }; // 55c8ab
    Poly pol(n + 1, e(pol)); pol[0] = e[1] = 1;
    for (++k; k; k /= 2) {
        if (k % 2) pol = combine(pol, e);
        e = combine(e, e);
    } // 8137be
    ll res = 0;
    rep(i,0,n) res = (res + pol[i + 1] * S[i]) % mod;
    return res;
} // 5948dc
```

**FloorModSum.h**  
**Description:** Floor sum and Mod sum.  
 $\text{modsum}(\text{to}, c, k, m) = \sum_{i=0}^{\text{to}-1} (ki + c) \% m$ .  $\text{divsum}$  is similar but for floored division.  
**Time:**  $\log(m)$ , with a large constant.

---

5c5bc5, 15L, 3m20s

```
typedef unsigned long long ull;
// written in a weird way to deal with overflows correctly
ull sumsq(ull to) { return to / 2 * ((to-1) | 1); }
ull divsum(ull to, ull c, ull k, ull m) {
    ull res = k / m * sumsq(to) + c / m * to;
    k %= m; c %= m;
    if (!k) return res;
    ull to2 = (to * k + c) / m;
    return res + (to - 1) * to2 - divsum(to2, m-1 - c, m, k);
} // 78bfc8
ll modsum(ull to, ll c, ll k, ll m) {
    c = ((c % m) + m) % m;
    k = ((k % m) + m) % m;
    return to * c + k * sumsq(to) - m * divsum(to, c, k, m);
} // 5daf3e
```

3.3 Linear Algebra

3.3.1 Matrices

**Determinant.h**  
**Description:** Calculates determinant of a matrix. Destroys the matrix.  
**Time:**  $\mathcal{O}(N^3)$

---

bd5cec, 15L, 2m30s

```
double det(vector<vector<double>>& a) {
    int n = sz(a); double res = 1;
    rep(i,0,n) {
        int b = i;
        rep(j,i+1,n) if (fabs(a[j][i]) > fabs(a[b][i])) b = j;
        if (i != b) swap(a[i], a[b]), res *= -1;
        res *= a[i][i];
        if (res == 0) return 0;
        rep(j,i+1,n) {
            double v = a[j][i] / a[i][i];
            if (v != 0) rep(k,i+1,n) a[j][k] -= v * a[i][k];
        } // 4ec6a2
    } // ee1466
    return res;
} // bd5cec
```

**IntDeterminant.h**  
**Description:** Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.  
**Time:**  $\mathcal{O}(N^3 \log \text{mod})$

---

3e3b26, 17L, 2m30s

```
ll det(vector<vector<ll>>& a, ll mod) {
    int n = sz(a); ll ans = 1;
    rep(i,0,n) {
        rep(j,i+1,n) {
            while (a[j][i] != 0) { // gcd step
                ll t = a[i][i] / a[j][i];
                if (t) rep(k,i,n)
                    a[i][k] = (a[i][k] - a[j][k] * t) % mod;
                swap(a[i], a[j]);
                ans *= -1;
            } // e81b29
        } // 30d1b2
        ans = ans * a[i][i] % mod;
        if (!ans) return 0;
    } // f39a45
    return (ans + mod) % mod;
} // 3e3b26
```

**MatrixInverse.h**  
**Description:** Invert matrix  $A$ . Returns rank; result is stored in  $A$  unless singular ( $\text{rank} < n$ ). Can easily be extended to prime moduli; for prime powers, repeatedly set  $A^{-1} = A^{-1}(2I - AA^{-1}) \pmod{p^k}$  where  $A^{-1}$  starts as the inverse of  $A \bmod p$ , and  $k$  is doubled in each step.  
**Time:**  $\mathcal{O}(n^3)$

---

ebfff6, 32L, 5m40s

```
int matInv(vector<vector<double>>& A) {
    int n = sz(A); vi col(n);
    vector<vector<double>> tmp(n, vector<double>(n));
    rep(i,0,n) tmp[i][i] = 1, col[i] = i;
    rep(i,0,n) {
        int r = i, c = i;
        rep(j,i,n) rep(k,i,n)
            if (fabs(A[j][k]) > fabs(A[r][c]))
                r = j, c = k;
        if (fabs(A[r][c]) < 1e-12) return i;
        A[i].swap(A[r]); tmp[i].swap(tmp[r]);
        rep(j,0,n)
            swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
        swap(col[i], col[c]);
        double v = A[i][i];
        rep(j,i+1,n) {
            double f = A[j][i] / v;
            A[j][i] = 0;
            rep(k,i+1,n) A[j][k] -= f*A[i][k];
            rep(k,0,n) tmp[j][k] -= f*tmp[i][k];
        } // ebcea3
        rep(j,i+1,n) A[i][j] /= v;
        rep(j,0,n) tmp[i][j] /= v;
        A[i][i] = 1;
    } // 26d90b
    for (int i = n-1; i > 0; --i) rep(j,0,i) {
        double v = A[j][i];
        rep(k,0,n) tmp[j][k] -= v*tmp[i][k];
    } // 03ae0c
    rep(i,0,n) rep(j,0,n) A[col[i]][col[j]] = tmp[i][j];
    return n;
} // ebfff6
```

3.3.2 Systems

**SolveLinear.h**  
**Description:** Solves  $A * x = b$ . If there are multiple solutions, an arbitrary one is returned. Returns rank, or -1 if no solutions. Data in  $A$  and  $b$  is lost.  
**Time:**  $\mathcal{O}(n^2 m)$

---

44c9ab, 46L, 7m30s

```
typedef vector<double> vd;
const double eps = 1e-12;
int solveLinear(vector<vd>& A, vd& b, vd& x) {
    int n = sz(A), m = sz(x), rank = 0, br, bc;
    if (n) assert(sz(A[0]) == m);
    vi col(m); iota(all(col), 0);
    rep(i,0,n) {
        double v, bv = 0;
        rep(r,i,n) rep(c,i,m)
            if ((v = fabs(A[r][c])) > bv)
                br = r, bc = c, bv = v;
        if (bv <= eps) {
            rep(j,i,n) if (fabs(b[j]) > eps) return -1;
            break;
        } // c92205
        swap(A[i], A[br]);
        swap(b[i], b[br]);
        swap(col[i], col[bc]);
        rep(j,0,n) swap(A[j][i], A[j][bc]);
        bv = 1/A[i][i];
        rep(j,i+1,n) {
            double fac = A[j][i] * bv;
            b[j] -= fac * b[i];
        }
    }
}
```

```
        rep(k,i+1,m) A[j][k] -= fac*A[i][k];
    } // 881860
    rank++;
} // 0f0f0f
x.assign(m, 0);
for (int i = rank; i--;) {
    b[i] /= A[i][i];
    x[col[i]] = b[i];
    rep(j,0,i) b[j] -= A[j][i] * b[i];
} // ed1d08
return rank; // (multiple solutions if rank < m)
} // ade67b
```

/\*  
To get all uniquely determined values of  $x$  back from  
SolveLinear, make the following changes:  
rep(j,0,n) if (j != i) // instead of rep(j,i+1,n)  
// ... then at the end:  
x.assign(m, undefined);  
rep(i,0,rank) {  
 rep(j,rank,m) if (fabs(A[i][j]) > eps) goto fail;  
 x[col[i]] = b[i] / A[i][i];  
fail:; } // 87878c  
\*/

SolveLinearBinary.h

**Description:** Solves  $Ax = b$  over  $\mathbb{F}_2$ . If there are multiple solutions, one is returned arbitrarily. Returns rank, or -1 if no solutions. Destroys  $A$  and  $b$ .  
**Time:**  $\mathcal{O}(n^2m)$

```
typedef bitset<1000> bs;
int solveLinear(vector<bs>& A, vi& b, bs& x, int m) {
    int n = sz(A), rank = 0, br;
    assert(m <= sz(x));
    vi col(m); iota(all(col), 0);
    rep(i,0,n) {
        for (br=i; br<n; ++br) if (A[br].any()) break;
        if (br == n) {
            rep(j,i,n) if(b[j]) return -1;
            break;
        } // 80687c
        int bc = (int)A[br]._Find_next(i-1);
        swap(A[i], A[br]);
        swap(b[i], b[br]);
        swap(col[i], col[bc]);
        rep(j,0,n) if (A[j][i] != A[j][bc]) {
            A[j].flip(i); A[j].flip(bc);
        } // b44a9b
        rep(j,i+1,n) if (A[j][i]) {
            b[j] ^= b[i];
            A[j] ^= A[i];
        } // 87192e
        rank++;
    } // fe9281
    x = bs();
    for (int i = rank; i--;) {
        if (!b[i]) continue;
        x[col[i]] = 1;
        rep(j,0,i) b[j] ^= A[j][i];
    } // 8fdbab
    return rank; // (multiple solutions if rank < m)
} // 26d73e
```

Tridiagonal.h

**Description:**  $x = \text{tridiagonal}(d, p, q, b)$  solves the equation system

$$\begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_{n-1} \end{pmatrix} = \begin{pmatrix} d_0 & p_0 & 0 & 0 & \cdots & 0 \\ q_0 & d_1 & p_1 & 0 & \cdots & 0 \\ 0 & q_1 & d_2 & p_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & q_{n-3} & d_{n-2} & p_{n-2} \\ 0 & 0 & \cdots & 0 & q_{n-2} & d_{n-1} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \end{pmatrix}.$$

This is useful for solving problems on the type

$$a_i = b_i a_{i-1} + c_i a_{i+1} + d_i, 1 \leq i \leq n,$$

where  $a_0, a_{n+1}, b_i, c_i$  and  $d_i$  are known.  $a$  can then be obtained from

$$\{a_i\} = \text{tridiagonal}(\{1, -1, -1, \dots, -1, 1\}, \{0, c_1, c_2, \dots, c_n\}, \{b_1, b_2, \dots, b_n, 0\}, \{a_0, d_1, d_2, \dots, d_n, a_{n+1}\}).$$

Fails if the solution is not unique.  
If  $|d_i| > |p_i| + |q_{i-1}|$  for all  $i$ , or  $|d_i| > |p_{i-1}| + |q_i|$ , or the matrix is positive definite, the algorithm is numerically stable and neither  $\text{tr}$  nor the check for  $\text{diag}[i] == 0$  is needed.  
**Time:**  $\mathcal{O}(N)$

```
typedef double T;
vector<T> tridiagonal(vector<T> diag, const vector<T>& super,
    const vector<T>& sub, vector<T> b) {
    int n = sz(b); vi tr(n);
    rep(i,0,n-1) {
        if (abs(diag[i]) < 1e-9 * abs(super[i])) { // diag[i] == 0
            b[i+1] -= b[i] * diag[i+1] / super[i];
            if (i+2 < n) b[i+2] -= b[i] * sub[i+1] / super[i];
            diag[i+1] = sub[i]; tr[++i] = 1;
        } else { // 464c09
            diag[i+1] -= super[i]*sub[i]/diag[i];
            b[i+1] -= b[i]*sub[i]/diag[i];
        } // 62de5a
    } // ed9cce
    for (int i = n; i--;) {
        if (tr[i]) {
            swap(b[i], b[i-1]);
            diag[i-1] = diag[i];
            b[i] /= super[i-1];
        } else { // 0543e4
            b[i] /= diag[i];
            if (i) b[i-1] -= b[i]*super[i-1];
        } // aa91c6
    } // 28af28
    return b;
} // 06d549
```

3.3.3 Optimization

Simplex.h

**Description:** Solves a general linear maximization problem: maximize  $c^T x$  subject to  $Ax \leq b, x \geq 0$ . Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of  $c^T x$  otherwise. The input vector is set to an optimal  $x$  (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that  $x = 0$  is viable.  
**Usage:** vvd A = {{1,-1}, {-1,1}, {-1,-2}};  
vd b = {1,1,-4}, c = {-1,-1}, x;  
T val = LPSolver(A, b, c).solve(x);  
**Time:**  $\mathcal{O}(NM * \# \text{pivots})$ , where a pivot may be e.g. an edge relaxation.  $\mathcal{O}(2^n)$  in the general case.

```
typedef double T; // long double, Rational, double + modP>...
typedef vector<T> vd;
typedef vector<vd> vvd;
const T eps = 1e-8, inf = 1/0.0;
#define MP make_pair
```

```
#define ltj(X) if(s == -1 || MP(X[j],N[j]) < MP(X[s],N[s])) s=j
struct LPSolver {
    int m, n; vi N, B; vvd D;
    LPSolver(const vvd& A, const vd& b, const vd& c) :
        m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
        rep(i,0,m) rep(j,0,n) D[i][j] = A[i][j];
        rep(i,0,m) { B[i] = n+i; D[i][n] = -1; D[i][n+1] = b[i]; }
        rep(j,0,n) { N[j] = j; D[m][j] = -c[j]; }
        N[n] = -1; D[m+1][n] = 1;
    } // 6ff8e9
    void pivot(int r, int s) {
        T *a = D[r].data(), inv = 1 / a[s];
        rep(i,0,m+2) if (i != r && abs(D[i][s]) > eps) {
            T *b = D[i].data(), inv2 = b[s] * inv;
            rep(j,0,n+2) b[j] -= a[j] * inv2;
            b[s] = a[s] * inv2;
        } // ca4460
        rep(j,0,n+2) if (j != s) D[r][j] *= inv;
        rep(i,0,m+2) if (i != r) D[i][s] *= -inv;
        D[r][s] = inv;
        swap(B[r], N[s]);
    } // 9cd0a8
    bool simplex(int phase) {
        int x = m + phase - 1;
        for (;;) {
            int s = -1;
            rep(j,0,n+1) if (N[j] != -phase) ltj(D[x]);
            if (D[x][s] >= -eps) return true;
            int r = -1;
            rep(i,0,m) {
                if (D[i][s] <= eps) continue;
                if (r == -1 || MP(D[i][n+1] / D[i][s], B[i])
                    < MP(D[r][n+1] / D[r][s], B[r])) r = i;
            } // 46853f
            if (r == -1) return false;
            pivot(r, s);
        } // 7d839b
    } // f15644
    T solve(vd &x) {
        int r = 0;
        rep(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
        if (D[r][n+1] < -eps) {
            pivot(r, n);
            if (!simplex(2) || D[m+1][n+1] < -eps) return -inf;
            rep(i,0,m) if (B[i] == -1) {
                int s = 0;
                rep(j,1,n+1) ltj(D[i]);
                pivot(i, s);
            } // 683310
        } // b6553f
        bool ok = simplex(1); x = vd(n);
        rep(i,0,m) if (B[i] < n) x[B[i]] = D[i][n+1];
        return ok ? D[m][n+1] : inf;
    } // 396a95
}; // c57b35
```

3.4 Numerical Calculus

3.4.1 Functions

Integrate.h

**Description:** Simple integration of a function over an interval using Simpson's rule. The error should be proportional to  $h^4$ , although in practice you will want to verify that the result is stable to desired precision when epsilon changes.

```
template<class F>
double quad(double a, double b, F f, const int n = 1000) {
    double h = (b - a) / 2 / n, v = f(a) + f(b);
    rep(i,1,n*2) v += f(a + i*h) * (i&1 ? 4 : 2);
}
```

```
    return v * h / 3;
} // ddcce2
```

IntegrateAdaptive.h

Description: Fast integration using an adaptive Simpson’s rule.

Usage: double sphereVolume = quad(-1, 1, [](double x) {  
return quad(-1, 1, [&](double y) {  
return quad(-1, 1, [&](double z) {  
return x\*x + y\*y + z\*z < 1; }));});});});  
}

92dd79, 12L, 2m50s

```
typedef double d;
#define S(a,b) (f(a) + 4*f((a+b) / 2) + f(b)) * (b-a) / 6
template <class F> d rec(F& f, d a, d b, d eps, d S) {
    d c = (a + b) / 2;
    d S1 = S(a, c), S2 = S(c, b), T = S1 + S2;
    if (abs(T - S) <= 15 * eps || b - a < 1e-10)
        return T + (T - S) / 15;
    return rec(f, a, c, eps / 2, S1) + rec(f, c, b, eps / 2, S2);
} // 10c680
template<class F> d quad(d a, d b, F f, d eps = 1e-8) {
    return rec(f, a, b, eps, S(a, b));
} // 7afc73
```

3.4.2 Polynomials

PolyRoots.h

Description: Finds the real roots to a polynomial. xmin and xmax should contain all real roots. Multiple roots may produce duplicate nearby answers; deduplicate if necessary.

Usage: polyRoots({{-2,-3,1}},-1e9,1e9) // solve x^2-3x+2 = 0

Time:  $O(n^2 \log(1/\epsilon))$

```
struct Poly {
    vector<double> a;
    double operator()(double x) const {
        double val = 0;
        for (int i = sz(a); i--;) (val += x) += a[i];
        return val;
    } // ae76f3
    void diff() {
        rep(i,1,sz(a)) a[i-1] = i*a[i];
        a.pop_back();
    } // afcaea
    void divroot(double x0) {
        double b = a.back(), c; a.back() = 0;
        for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+b, b=c;
        a.pop_back();
    } // 3f874a
}; // c9b7b0
vector<double> polyRoots(Poly p, double xmin, double xmax) {
    if (sz(p.a) == 2) { return {-p.a[0]/p.a[1]}; }
    vector<double> ret;
    Poly der = p;
    der.diff();
    auto dr = polyRoots(der, xmin, xmax);
    dr.push_back(xmin-1);
    dr.push_back(xmax+1);
    sort(all(dr));
    rep(i,0,sz(dr)-1) {
        double l = dr[i], h = dr[i+1];
        bool sign = p(l) > 0;
        if (sign ^ (p(h) > 0)) {
            rep(it,0,60) { // while (h - l > 1e-8)
                double m = (l + h) / 2, f = p(m);
                if ((f <= 0) ^ sign) l = m;
                else h = m;
            } // b69f41
            ret.push_back((l + h) / 2);
        } // fc22f0
    }
```

```
    } // d15986
    return ret;
} // b00bfe
```

PolyInterpolate.h

Description: Given  $n$  points  $(x[i], y[i])$ , computes an  $n-1$ -degree polynomial  $p$  that passes through them:  $p(x) = a[0] * x^0 + \dots + a[n-1] * x^{n-1}$ . For numerical precision, pick  $x[k] = c * \cos(k/(n-1) * \pi), k = 0 \dots n-1$ .

Time:  $O(n^2)$

```
typedef vector<double> vd;
vd interpolate(vd x, vd y, int n) {
    vd res(n), temp(n);
    rep(k,0,n-1) rep(i,k+1,n)
        y[i] = (y[i] - y[k]) / (x[i] - x[k]);
    double last = 0; temp[0] = 1;
    rep(k,0,n) rep(i,0,n) {
        res[i] += y[k] * temp[i];
        swap(last, temp[i]);
        temp[i] -= last * x[k];
    } // 4c74fe
    return res;
} // 285367
```

Geometry (4)

4.1 Geometric primitives

Point.h

Description: Class to handle points in the plane. T can be e.g. double or long long. (Avoid int.)

```
template <class T> int sgn(T x) { return (x > 0) - (x < 0); }
template<class T>
struct Point {
    typedef Point P;
    T x, y;
    explicit Point(T _x=0, T _y=0) : x(_x), y(_y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    P operator+(P p) const { return P(x+p.x, y+p.y); }
    P operator-(P p) const { return P(x-p.x, y-p.y); }
    P operator*(T d) const { return P(x*d, y*d); }
    P operator/(T d) const { return P(x/d, y/d); }
    T dot(P p) const { return x*p.x + y*p.y; }
    T cross(P p) const { return x*p.y - y*p.x; }
    T cross(P a, P b) const { return (a-*this).cross(b-*this); }
    T dist2() const { return x*x + y*y; }
    double dist() const { return sqrt((double)dist2()); }
    // angle to x-axis in interval [-pi, pi]
    double angle() const { return atan2(y, x); }
    P unit() const { return *this/dist(); } // makes dist()==1
    P perp() const { return P(-y, x); } // rotates +90 degrees
    P normal() const { return perp().unit(); }
    // returns point rotated 'a' radians ccw around the origin
    P rotate(double a) const {
        return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); } // 4822a3
    friend ostream& operator<<(ostream& os, P p) {
        return os << "(" << p.x << ", " << p.y << ")"; } // 9a9c95
}; // 6a5cce
```

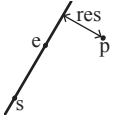
lineDistance.h

Description:

Returns the signed distance between point p and the line containing points a and b. Positive value on left side and negative on right as seen from a towards b. a==b gives nan. P is supposed to be Point<T> or Point3D<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long. Using Point3D will always give a non-negative distance. For Point3D, call .dist on the result of the cross product.

"Point.h"

f6bf6b, 4L, 50s



```
template<class P>
double lineDist(const P& a, const P& b, const P& p) {
    return (double) (b-a).cross(p-a) / (b-a).dist();
} // 00891c
```

SegmentDistance.h

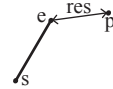
Description:

Returns the shortest distance between point p and the line segment from point s to e.

Usage: Point<double> a, b(2,2), p(1,1);  
bool onSegment = segDist(a,b,p) < 1e-10;

"Point.h"

5c88f4, 6L, 1m20s



```
typedef Point<double> P;
double segDist(P& s, P& e, P& p) {
    if (s==e) return (p-s).dist();
    auto d = (e-s).dist2(), t = min(d,max(.0, (p-s).dot(e-s)));
    return ((p-s)*d-(e-s)*t).dist()/d;
} // ae751a
```

SegmentIntersection.h

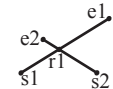
Description:

If a unique intersection point between the line segments going from s1 to e1 and from s2 to e2 exists then it is returned. If no intersection point exists an empty vector is returned. If infinitely many exist a vector with 2 elements is returned, containing the endpoints of the common line segment. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or long long.

Usage: vector<P> inter = segInter(s1,e1,s2,e2);  
if (sz(inter)==1)  
cout << "segments intersect at " << inter[0] << endl;

"Point.h", "OnSegment.h"

9d57f2, 13L, 3m10s

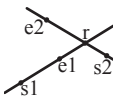


```
template<class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b),
        oc = a.cross(b, c), od = a.cross(b, d);
    // Checks if intersection is single non-endpoint point.
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSegment(c, d, a)) s.insert(a);
    if (onSegment(c, d, b)) s.insert(b);
    if (onSegment(a, b, c)) s.insert(c);
    if (onSegment(a, b, d)) s.insert(d);
    return {all(s)};
} // 9d57f2
```

lineIntersection.h

Description:

If a unique intersection point of the lines going through s1,e1 and s2,e2 exists {1, point} is returned. If no intersection point exists {0, (0,0)} is returned and if infinitely many exists {-1, (0,0)} is returned. The wrong position will be returned if P is Point<ll> and the intersection point does not have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow if using int or ll.



```
Usage: auto res = lineInter(s1,e1,s2,e2);
if (res.first == 1)
cout << "intersection point at " << res.second << endl;
"Point.h"
a01f81, 8L, 1m50s
template<class P>
pair<int, P> lineInter(P s1, P e1, P s2, P e2) {
    auto d = (e1 - s1).cross(e2 - s2);
    if (d == 0) // if parallel
        return {-(s1.cross(e1, s2) == 0), P(0, 0)};
    auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
    return {1, (s1 * p + e1 * q) / d};
} // 47279a
```

```
sideOf.h
Description: Returns where p is as seen from s towards e. 1/0/-1 ⇔ left/on line/right. If the optional argument eps is given 0 is returned if p is within distance eps from the line. P is supposed to be Point<T> where T is e.g. double or long long. It uses products in intermediate steps so watch out for overflow if using int or long long.
Usage: bool left = sideOf(p1,p2,q)==1;
"Point.h"
3af81c, 9L, 1m40s
template<class P>
int sideOf(P s, P e, P p) { return sgn(s.cross(e, p)); }
```

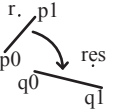
```
template<class P>
int sideOf(const P& s, const P& e, const P& p, double eps) {
    auto a = (e-s).cross(p-s);
    double l = (e-s).dist()*eps;
    return (a > l) - (a < -l);
} // 33fa03
```

```
OnSegment.h
Description: Returns true iff p lies on the line segment from s to e. Use (segDist(s,e,p)<=epsilon) instead when using Point<double>.
"Point.h"
c597e8, 3L, 40s
```

```
template<class P> bool onSegment(P s, P e, P p) {
    return p.cross(s, e) == 0 && (s - p).dot(e - p) <= 0;
} // c597e8
```

linearTransformation.h

Description:



```
"Point.h"
03a306, 6L, 1m40s
typedef Point<double> P;
P linearTransformation(const P& p0, const P& p1,
    const P& q0, const P& q1, const P& r) {
    P dp = p1-p0, dq = q1-q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + P((r-p0).cross(num), (r-p0).dot(num))/dp.dist2();
} // 45ea01
```

LineProjectionReflection.h

```
Description: Projects point p onto line ab. Set refl=true to get reflection of point p across line ab instead. The wrong point will be returned if P is an integer point and the desired point doesn't have integer coordinates. Products of three coordinates are used in intermediate steps so watch out for overflow.
"Point.h"
b5562d, 5L, 50s
template<class P>
P lineProj(P a, P b, P p, bool refl=false) {
    P v = b - a;
    return p - v.perp()*(1+refl)*v.cross(p-a)/v.dist2();
} // 4b716b
```

Angle.h

Description: A class for ordering angles (as represented by int points and a number of rotations around the origin). Useful for rotational sweeping. Sometimes also represents points or vectors.

```
Usage: vector<Angle> v = {w[0], w[0].t360() ...}; // sorted
int j = 0; rep(i,0,n) { while (v[j] < v[i].t180()) ++j; }
sweeps j such that (j-i) represents the number of positively oriented triangles with vertices at 0 and i
6b951c, 35L, 7m50s
struct Angle {
    ll x, y;
    int t;
    Angle(ll x, ll y, int t=0) : x(x), y(y), t(t) {}
    Angle operator-(Angle b) const { return {x-b.x, y-b.y, t}; }
    int half() const {
        assert(x || y);
        return y < 0 || (y == 0 && x < 0);
    } // c935fb
    Angle t90() const { return {-y, x, t + (half() && x >= 0)}; }
    Angle t180() const { return {-x, -y, t + half()}; }
    Angle t360() const { return {x, y, t + 1}; }
}; // 88ad55
bool operator<(Angle a, Angle b) {
    // add a.dist2() and b.dist2() to also compare distances
    return make_tuple(a.t, a.half(), a.y * b.x) <
        make_tuple(b.t, b.half(), a.x * b.y);
} // 9c58ca
```

```
// Given two points, this calculates the smallest angle between
// them, i.e., the angle that covers the defined line segment.
pair<Angle, Angle> segmentAngles(Angle a, Angle b) {
    if (b < a) swap(a, b);
    return (b < a.t180() ?
        make_pair(a, b) : make_pair(b, a.t360()));
} // 5eac29
Angle operator+(Angle a, Angle b) { // point a + vector b
    Angle r(a.x + b.x, a.y + b.y, a.t);
    if (a.t180() < r) r.t--;
    return r.t180() < a ? r.t360() : r;
} // 3d8073
Angle angleDiff(Angle a, Angle b) { // angle b - angle a
    int tu = b.t - a.t; a.t = b.t;
    return {a.x*b.x + a.y*b.y, a.x*b.y - a.y*b.x, tu - (b < a)};
} // ba3082
```

4.2 Circles

CircleIntersection.h

```
Description: Computes the pair of points at which two circles intersect. Returns false in case of no intersection.
"Point.h"
84d6d3, 11L, 2m50s
typedef Point<double> P;
bool circleInter(P a,P b,double r1,double r2,pair<P, P>* out) {
    if (a == b) { assert(r1 != r2); return false; }
    P vec = b - a;
    double d2 = vec.dist2(), sum = r1+r2, dif = r1-r2,
        p = (d2 + r1*r1 - r2*r2)/(d2*2), h2 = r1*r1 - p*p*d2;
    if (sum*sum < d2 || dif*dif > d2) return false;
    P mid = a + vec*p, per = vec.perp() * sqrt(fmax(0, h2) / d2);
    *out = {mid + per, mid - per};
    return true;
} // c64785
```

CircleTangents.h

Description: Finds the external tangents of two circles, or internal if r2 is negated. Can return 0, 1, or 2 tangents – 0 if one circle contains the other (or overlaps it, in the internal case, or if the circles are the same); 1 if the circles are tangent to each other (in which case .first = .second and the tangent line is perpendicular to the line between the centers). .first and .second give the tangency points at circle 1 and 2 respectively. To find the tangents of a circle with a point set r2 to 0.

```
"Point.h"
b0153d, 13L, 2m40s
template<class P>
vector<pair<P, P>> tangents(P c1, double r1, P c2, double r2) {
    P d = c2 - c1;
    double dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 == 0 || h2 < 0) return {};
    vector<pair<P, P>> out;
    for (double sign : {-1, 1}) {
        P v = (d * dr + d.perp() * sqrt(h2) * sign) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    } // e25263
    if (h2 == 0) out.pop_back();
    return out;
} // 4835b9
```

CircleLine.h

Description: Finds the intersection between a circle and a line. Returns a vector of either 0, 1, or 2 intersection points. P is intended to be Point<double>.

```
"Point.h"
e0cfba, 9L, 1m50s
template<class P>
vector<P> circleLine(P c, double r, P a, P b) {
    P ab = b - a, p = a + ab * (c-a).dot(ab) / ab.dist2();
    double s = a.cross(b, c), h2 = r*r - s*s / ab.dist2();
    if (h2 < 0) return {};
    if (h2 == 0) return {p};
    P h = ab.unit() * sqrt(h2);
    return {p - h, p + h};
} // 59aded
```

CirclePolygonIntersection.h

Description: Returns the area of the intersection of a circle with a ccw polygon.

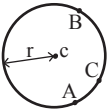
```
Time: O(n)
".../content/geometry/Point.h"
19add1, 19L, 4m10s
typedef Point<double> P;
#define arg(p, q) atan2(p.cross(q), p.dot(q))
double circlePoly(P c, double r, vector<P> ps) {
    auto tri = [&](P p, P q) {
        auto r2 = r * r / 2;
        P d = q - p;
        auto a = d.dot(p)/d.dist2(), b = (p.dist2()-r*r)/d.dist2();
        auto det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        auto s = max(0., -a-sqrt(det)), t = min(1., -a+sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = q + d * (t-1);
        return arg(p,u) * r2 + u.cross(v)/2 + arg(v,q) * r2;
    }; // a526fe
    auto sum = 0.0;
    rep(i,0,sz(ps))
        sum += tri(ps[i] - c, ps[(i + 1) % sz(ps)] - c);
    return sum;
} // f082e0
```



### circumcircle.h

Description:

The circumrle of a triangle is the circle intersecting all three vertices. ccRadius returns the radius of the circle going through points A, B and C and ccCenter returns the center of the same circle.



|                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                    | 1caa3aa, 9L, 1m50s |
| <b>typedef</b> Point<double> P;<br><b>double</b> ccRadius( <b>const</b> P& A, <b>const</b> P& B, <b>const</b> P& C) {<br><b>return</b> (B-A).dist()* (C-B).dist()* (A-C).dist() /<br>abs((B-A).cross(C-A)) / 2;<br>}<br>// 607d98<br>P ccCenter( <b>const</b> P& A, <b>const</b> P& B, <b>const</b> P& C) {<br>P b = C-A, c = B-A;<br><b>return</b> A + (b*c.dist2()-c*b.dist2()).perp() / b.cross(c) / 2;<br>}<br>// 79372e |                    |

### MinimumEnclosingCircle.h

Description: Computes the minimum circle that encloses a set of points.

Time: expected  $\mathcal{O}(n)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| "circumcircle.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 09dd0aa, 17L, 3m |
| pair<P, double> mec(vector<P> ps) {<br>shuffle(all(ps), mt19937(time(0)));<br>P o = ps[0];<br>double r = 0, EPS = 1 + 1e-8;<br>rep(i,0,sz(ps)) <b>if</b> ((o - ps[i]).dist() > r * EPS) {<br>o = ps[i], r = 0;<br>rep(j,0,i) <b>if</b> ((o - ps[j]).dist() > r * EPS) {<br>o = (ps[i] + ps[j]) / 2;<br>r = (o - ps[i]).dist();<br>rep(k,0,j) <b>if</b> ((o - ps[k]).dist() > r * EPS) {<br>o = ccCenter(ps[i], ps[j], ps[k]);<br>r = (o - ps[i]).dist();<br>}<br>}<br>}<br>// 7b0ecf<br>// dcf00e<br><b>return</b> {o, r};<br>}<br>// 09dd0aa |                  |

## 4.3 Polygons

### InsidePolygon.h

Description: Returns true if p lies within the polygon. If strict is true, it returns false for points on the boundary. The algorithm uses products in intermediate steps so watch out for overflow.

Usage: vector<P> v = {P{4,4}, P{1,2}, P{2,1}};

bool in = inPolygon(v, P{3, 3}, false);

Time:  $\mathcal{O}(n)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "Point.h", "OnSegment.h", "SegmentDistance.h"                                                                                                                                                                                                                                                                                                                                                                                                    | 2bf504, 11L, 2m10s |
| <b>template&lt;class</b> P<br><b>bool</b> inPolygon(vector<P> &p, P a, <b>bool</b> strict = <b>true</b> ) {<br><b>int</b> cnt = 0, n = sz(p);<br>rep(i,0,n) {<br>P q = p[(i + 1) % n];<br><b>if</b> (onSegment(p[i], q, a)) <b>return</b> !strict;<br>//or: <b>if</b> (segDist(p[i], q, a) <= eps) <b>return</b> !strict;<br>cnt ^= ((a.y<p[i].y) - (a.y<q.y)) * a.cross(p[i], q) > 0;<br>}<br>// 1b9961<br><b>return</b> cnt;<br>}<br>// c7225e |                    |

### PolygonCenter.h

Description: Returns the center of mass for a polygon.

Time:  $\mathcal{O}(n)$

|                                 |                   |
|---------------------------------|-------------------|
| "Point.h"                       | 9706dc, 9L, 1m40s |
| <b>typedef</b> Point<double> P; |                   |

|                                                                                                                                                                                                                                                                                                        |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| P polygonCenter( <b>const</b> vector<P>& v) {<br>P res(0, 0); <b>double</b> A = 0;<br><b>for</b> ( <b>int</b> i = 0, j = sz(v) - 1; i < sz(v); j = i++) {<br>res = res + (v[i] + v[j]) * v[j].cross(v[i]);<br>A += v[j].cross(v[i]);<br>}<br>// 307102<br><b>return</b> res / A / 3;<br>}<br>// 0d0d84 |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

### PolygonCut.h

Description:

Returns a vector with the vertices of a polygon with every-thing to the left of the line going from s to e cut away.

Usage: vector<P> p = ...;

p = polygonCut(p, P(0,0), P(1,0));

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                   | d07181, 13L, 2m20s |
| <b>typedef</b> Point<double> P;<br>vector<P> polygonCut( <b>const</b> vector<P>& poly, P s, P e) {<br>vector<P> res;<br>rep(i,0,sz(poly)) {<br>P cur = poly[i], prev = i ? poly[i-1] : poly.back();<br><b>auto</b> a = s.cross(e, cur), b = s.cross(e, prev);<br><b>if</b> ((a < 0) != (b < 0))<br>res.push_back(cur + (prev - cur) * (a / (a - b)));<br><b>if</b> (a < 0)<br>res.push_back(cur);<br>}<br>// 757c0d<br><b>return</b> res;<br>}<br>// 42c993 |                    |

### PolygonUnion.h

Description: Calculates the area of the union of  $n$  polygons (not necessarily convex). The points within each polygon must be given in CCW order. (Epsilon checks may optionally be added to sideOf/sgn, but shouldn't be needed.)

Time:  $\mathcal{O}(N^2)$ , where  $N$  is the total number of points

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "Point.h", "sideOf.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3931c6, 33L, 7m10s |
| <b>typedef</b> Point<double> P;<br><b>double</b> rat(P a, P b) { <b>return</b> sgn(b.x) ? a.x/b.x : a.y/b.y; }<br><b>double</b> polyUnion(vector<vector<P>>& poly) {<br><b>double</b> ret = 0;<br>rep(i,0,sz(poly)) rep(v,0,sz(poly[i])) {<br>P A = poly[i][v], B = poly[i][(v + 1) % sz(poly[i])];<br>vector<pair< <b>double</b> , <b>int</b> >> segs = {{0, 0}, {1, 0}};<br>rep(j,0,sz(poly)) <b>if</b> (i != j) {<br>rep(u,0,sz(poly[j])) {<br>P C = poly[j][u], D = poly[j][(u + 1) % sz(poly[j])];<br><b>int</b> sc = sideOf(A, B, C), sd = sideOf(A, B, D);<br><b>if</b> (sc != sd) {<br><b>double</b> sa = C.cross(D, A), sb = C.cross(D, B);<br><b>if</b> (min(sc, sd) < 0)<br>segs.emplace_back(sa / (sa - sb), sgn(sc - sd));<br><b>else if</b> (!sc && !sd && j<i && sgn((B-A).dot(D-C))>0) {<br>// cf73dd<br>segs.emplace_back(rat(C - A, B - A), 1);<br>segs.emplace_back(rat(D - A, B - A), -1);<br>}<br>}<br>}<br>// fdc855<br>sort(all(segs));<br><b>for</b> ( <b>auto</b> & s : segs) s.first = min(max(s.first, 0.0), 1.0);<br><b>double</b> sum = 0;<br><b>int</b> cnt = segs[0].second;<br>rep(j,1,sz(segs)) {<br><b>if</b> (!cnt) sum += segs[j].first - segs[j - 1].first;<br>cnt += segs[j].second;<br>}<br>// f5884f |                    |

|                                                                                        |  |
|----------------------------------------------------------------------------------------|--|
| ret += A.cross(B) * sum;<br>}<br>// 19180c<br><b>return</b> ret / 2;<br>}<br>// 6e88cf |  |
|----------------------------------------------------------------------------------------|--|

### ConvexHull.h

Description:

Returns a vector of the points of the convex hull in counter-clockwise order. Points on the edge of the hull between two other points are not considered part of the hull. To include these "borders", follow the code comments.

Time:  $\mathcal{O}(n \log n)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 310954, 13L, 2m50s |
| <b>typedef</b> Point<ll> P;<br>vector<P> convexHull(vector<P> pts) {<br><b>if</b> (sz(pts) <= 1) <b>return</b> pts;<br>sort(all(pts));<br>vector<P> h(sz(pts)+1); // 2*sz(pts) to include borders<br><b>int</b> s = 0, t = 0;<br><b>for</b> ( <b>int</b> it = 2; it--; s = --t, reverse(all(pts)))<br><b>for</b> (P p : pts) {<br><b>while</b> (t >= s + 2 && h[t-2].cross(h[t-1], p) <= 0) t--;<br>h[t++] = p; // < above to include borders<br>}<br>// bf0344<br><b>return</b> {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};<br>}<br>// ec85f8 |                    |

### HullDiameter.h

Description: Returns the two points with max distance on a convex hull (ccw, no duplicate/collinear points).

Time:  $\mathcal{O}(n)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                       |                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                             | c571b8, 12L, 2m20s |
| <b>typedef</b> Point<ll> P;<br>array<P, 2> hullDiameter(vector<P> S) {<br><b>int</b> n = sz(S), j = n < 2 ? 0 : 1;<br>pair<ll, array<P, 2>> res({0, {S[0], S[0]}});<br>rep(i,0,j)<br><b>for</b> (; j = (j + 1) % n) {<br>res = max(res, {{S[i] - S[j]}.dist2(), {S[i], S[j]}});<br><b>if</b> ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)<br><b>break</b> ;<br>}<br>// 56cc40<br><b>return</b> res.second;<br>}<br>// 5f726b |                    |

### PointInsideHull.h

Description: Determine whether a point t lies inside a convex hull (CCW order, with no collinear points). Returns true if point lies within the hull. If strict is true, points on the boundary aren't included.

Time:  $\mathcal{O}(\log N)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "Point.h", "sideOf.h", "OnSegment.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 71446b, 14L, 3m |
| <b>typedef</b> Point<ll> P;<br><br><b>bool</b> inHull( <b>const</b> vector<P>& l, P p, <b>bool</b> strict = <b>true</b> ) {<br><b>int</b> a = 1, b = sz(l) - 1, r = !strict;<br><b>if</b> (sz(l) < 3) <b>return</b> r && onSegment(l[0], l.back(), p);<br><b>if</b> (sideOf(l[0], l[a], l[b]) > 0) swap(a, b);<br><b>if</b> (sideOf(l[0], l[a], p) >= r    sideOf(l[0], l[b], p) <= -r)<br><b>return</b> false;<br><b>while</b> (abs(a - b) > 1) {<br><b>int</b> c = (a + b) / 2;<br>(sideOf(l[0], l[c], p) > 0 ? b : a) = c;<br>}<br>// b265ab<br><b>return</b> sgn(l[a].cross(l[b], p)) < r;<br>}<br>// c74639 |                 |

LineHullIntersection.h

**Description:** Line-convex polygon intersection. The polygon must be ccw and have no collinear points. lineHull(line, poly) returns a pair describing the intersection of a line with the polygon:  $\bullet(-1, -1)$  if no collision,  $\bullet(i, -1)$  if touching the corner  $i$ ,  $\bullet(i, i)$  if along side  $(i, i+1)$ ,  $\bullet(i, j)$  if crossing sides  $(i, i+1)$  and  $(j, j+1)$ . In the last case, if a corner  $i$  is crossed, this is treated as happening on side  $(i, i+1)$ . The points are returned in the same order as the line hits the polygon. extrVertex returns the point of a hull with the max projection onto a line.  
**Time:**  $\mathcal{O}(\log n)$

```
"Point.h" 7cf45b, 38L, 8m

#define cmp(i,j) sgn(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i+1, i) >= 0 && cmp(i, i-1+n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
    } // 68a24c
    return lo;
} // 7f0477
#define cmpL(i) sgn(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0)
        return {-1, -1};
    array<int, 2> res;
    rep(i,0,2) {
        int lo = endB, hi = endA, n = sz(poly);
        while ((lo + 1) % n != hi) {
            int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
            (cmpL(m) == cmpL(endB) ? lo : hi) = m;
        } // 52528c
        res[i] = (lo + !cmpL(hi)) % n;
        swap(endA, endB);
    } // c05c70
    if (res[0] == res[1]) return {res[0], -1};
    if (!cmpL(res[0]) && !cmpL(res[1]))
        switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
            case 0: return {res[0], res[0]};
            case 2: return {res[1], res[1]};
        } // 8fa383
    return res;
} // 36fc8e
```

PolygonContainmentTree.h

**Description:** building tree of polygon containment  
**Memory:**  $\mathcal{O}(N)$   
**Time:**  $\mathcal{O}(N \log N)$

```
14c82a, 44L, 8m40s

struct P { ll x, y; };

int current_x;
struct Segment {
    int idx; P p1, p2; bool is_upper;
    Segment(P p, P q, int i): idx(i), p1(p), p2(q), is_upper(p2.x < p1.x) { if (is_upper)swap(p1, p2); }
    ld get_y(ll x) const { return (ld) (p2.y - p1.y) / (p2.x - p1.x) * (x - p1.x) + p1.y; }
    tuple<ld, bool, int> get_comp() const { return {get_y(current_x), is_upper, p2.x}; }
    bool operator<(const Segment & o) const { return get_comp() < o.get_comp(); }
}; // fc8b4f
```

```
vector<int> build(vector<vector<P>>& polygons) {
    int n = sz(polygons);
    vector<tuple<int, int, int, Segment>> edges; // polygon edges
    rep(idx, 0, n) {
        const auto & v = polygons[idx];
        rep(i, 0, sz(v)) {
            int j = (i + 1) % sz(v);
            if (v[i].x == v[j].x)continue; // ignores vertical edges
            Segment seg = Segment(v[i], v[j], idx);
            edges.eb(seg.p1.x, 0, -seg.p1.y, seg);
            edges.eb(seg.p2.x, 1, -seg.p2.y, seg);
        } // b28b76
    } // 2603da
    sort(edges.begin(), edges.end());
    set<Segment> s;
    vector pai(n+1, n), vis(n, 0);
    for (auto [l, t, y, seg]: edges) {
        current_x = l;
        int i = seg.idx;
        if (t == 0) {
            if (not vis[i]) {
                vis[i] = true;
                auto it = s.upper_bound(seg);
                if (it == s.end())pai[i] = n;
                else if (it->is_upper)pai[i] = it->idx;
                else pai[i] = pai[it->idx];
            } // 17d107
            s.insert(seg);
        } // aff5b0
        else s.erase(seg);
    } // 3514ff
    return pai;
} // 0b7cbd
```

Minkowski.h

**Description:** Calculates the Minkowski sum of two convex Polygons in ccw order  
**Time:**  $\mathcal{O}(N + M)$

```
"Point.h" e510cd, 19L, 4m10s

template<typename P>
vector<P> minkowski(vector<P> p, vector<P> q) {
    auto comp = [](auto a, auto b) {
        return make_pair(a.y, a.x) < make_pair(b.y, b.x);
    }; // 278e98
    rotate(p.begin(), min_element(all(p), comp), p.end());
    rotate(q.begin(), min_element(all(q), comp), q.end());
    p.pb(p[0]); p.pb(p[1]);
    q.pb(q[0]); q.pb(q[1]);
    vector<P> result;
    int i = 0, j = 0;
    while(i < sz(p) - 2 || j < sz(q) - 2){
        result.pb(p[i] + q[j]);
        auto cross = (p[i + 1] - p[i]).cross(q[j + 1] - q[j]);
        if(cross >= 0 && i < sz(p) - 2) i++;
        if(cross <= 0 && j < sz(q) - 2) j++;
    } // 57f2d7
    return result;
} // a99371
```

HalfPlane.h

**Description:** Calculates the halfplane intersection of Halfplanes, Halfplanes allow everything to the left. Be careful with precision. Intersects with an artificial  $[-INF, INF]^2$  bounding box.  
**Time:**  $\mathcal{O}(N \log N)$

```
"Point.h" 0a6375, 50L, 10m40s
const ld EPS = 1e-9, INF = 1e9;
```

```
typedef Point<ld> P;
struct HP {
    P p, pq;
    ld angle;
    HP() {}
    HP(P a, P b) : p(a), pq(b - a), angle(atan2(pq.y, pq.x)) {}

    bool out(const P & r) { return pq.cross(r - p) < -EPS; }
    bool operator<(const HP & e) const { return angle < e.angle; }

    friend P inter(const HP & s, const HP & t) {
        ld alpha = (t.p - s.p).cross(t.pq) / s.pq.cross(t.pq);
        return s.p + (s.pq * alpha);
    } // 946b1a
}; // 782415
vector<P> hp_intersect(vector<HP> H) {
    P box[4] = {
        P(INF, INF),
        P(-INF, INF),
        P(-INF, -INF),
        P(INF, -INF)
    }; // 63b3ab
    for(int i = 0; i < 4; i++) H.eb(box[i], box[(i+1) % 4]);
    sort(H.begin(), H.end());
    deque<HP> dq;
    int len = 0;
    rep(i, 0, sz(H)) {
        while (len > 1 && H[i].out(inter(dq[len-1], dq[len-2]))
            )
            dq.pop_back(), len--;
        while (len > 1 && H[i].out(inter(dq[0], dq[1])))
            dq.pop_front(), len--;
        // Special case check: Parallel half planes
        if (len > 0 && fabs1(H[i].pq.cross(dq[len-1].pq)) < EPS
            ) {
            if (H[i].pq.dot(dq[len-1].pq) < 0.0) return {};
            if (H[i].out(dq[len-1].p))
                dq.pop_back(), len--;
            else continue;
        } // 6ef632
        dq.push_back(H[i]); len++;
    } // 82e81e
    while (len > 2 && dq[0].out(inter(dq[len-1], dq[len-2])))
        dq.pop_back(), len--;
    while (len > 2 && dq[len-1].out(inter(dq[0], dq[1])))
        dq.pop_front(), len--;
    if (len < 3) return {};
    vector<P> ret(len);
    rep(i, 0, len-1) ret[i] = inter(dq[i], dq[i+1]);
    ret.back() = inter(dq[len-1], dq[0]);
    return ret;
} // 8ab567
```

4.4 Misc. Point Set Problems

ClosestPair.h

**Description:** Finds the closest pair of points.  
**Time:**  $\mathcal{O}(n \log n)$

```
"Point.h" ac41a6, 17L, 3m10s

typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
    assert(sz(v) > 1);
    set<P> S;
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d(1 + (ll)sqrt(ret.first), 0);
        while (v[j].y <= p.y - d.x S.erase(v[j++]);
```

```
    auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
    for (; lo != hi; ++lo)
        ret = min(ret, {( *lo - p).dist2(), { *lo, p } });
    S.insert(p);
} // 5b096c
return ret.second;
} // bf22c6
```

ManhattanMST.h

**Description:** Given N points, returns up to 4\*N edges, which are guaranteed to contain a minimum spanning tree for the graph with edge weights  $w(p, q) = -p.x - q.x - p.y - q.y$ . Edges are in the form (distance, src, dst). Use a standard MST algorithm on the result to find the final MST. **Time:**  $\mathcal{O}(N \log N)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ee6376, 23L, 4m |
| <pre>typedef Point&lt;ll&gt; P; vector&lt;array&lt;ll, 3&gt;&gt; manhattanMST(vector&lt;P&gt; ps) {     vi id(sz(ps));     iota(all(id), 0);     vector&lt;array&lt;ll, 3&gt;&gt; edges;     rep(k,0,4) {         sort(all(id), [&amp;](int i, int j) {             return (ps[i]-ps[j]).x &lt; (ps[j]-ps[i]).y;}); // 02bd87         map&lt;ll, int&gt; sweep;         for (int i : id) {             for (auto it = sweep.lower_bound(-ps[i].y);                 it != sweep.end(); sweep.erase(it++)) {                 int j = it-&gt;second;                 P d = ps[i] - ps[j];                 if (d.y &gt; d.x) break;                 edges.push_back({d.y + d.x, i, j});             } // 271a42             sweep[-ps[i].y] = i;         } // e692f6         for (P&amp; p : ps) if (k &amp; 1) p.x = -p.x; else swap(p.x, p.y);     } // efbf43     return edges; } // babb49</pre> |                 |

FastDelaunay.h

**Description:** Fast Delaunay triangulation. Each circumcircle contains none of the input points. There must be no duplicate points. If all points are on a line, no triangles will be returned. Should work for doubles as well, though there may be precision issues in 'circ'. Returns triangles in order {t[0][0], t[0][1], t[0][2], t[1][0], ... }, all counter-clockwise. **Time:**  $\mathcal{O}(n \log n)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| "Point.h"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | eefdf5, 82L, 17m10s |
| <pre>typedef Point&lt;ll&gt; P; typedef struct Quad* Q; typedef __int128_t lll; // (can be ll if coords are &lt; 2e4) P arb(LLONG_MAX,LLONG_MAX); // not equal to any other point struct Quad {     Q rot, o; P p = arb; bool mark;     P&amp; F() { return r()-&gt;p; }     Q&amp; r() { return rot-&gt;rot; }     Q prev() { return rot-&gt;o-&gt;rot; }     Q next() { return r()-&gt;prev(); } } *H; // 18059e bool circ(P p, P a, P b, P c) { // is p in the circumcircle?     lll p2 = p.dist2(), A = a.dist2()-p2,         B = b.dist2()-p2, C = c.dist2()-p2;     return p.cross(a,b)*C + p.cross(b,c)*A + p.cross(c,a)*B &gt; 0; } // 6aff7b Q makeEdge(P orig, P dest) {     Q r = H ? H : new Quad{new Quad{new Quad{0}}};     H = r-&gt;o; r-&gt;r()-&gt;r() = r;     rep(i,0,4) r = r-&gt;rot, r-&gt;p = arb, r-&gt;o = i &amp; 1 ? r : r-&gt;r();</pre> |                     |

```
    r->p = orig; r->F() = dest;
    return r;
} // b3b5b1
void splice(Q a, Q b) {
    swap(a->o->rot->o, b->o->rot->o); swap(a->o, b->o);
} // 86ce01
Q connect(Q a, Q b) {
    Q q = makeEdge(a->F(), b->p);
    splice(q, a->next());
    splice(q->r(), b);
    return q;
} // 4a4fc2
pair<Q,Q> rec(const vector<P>& s) {
    if (sz(s) <= 3) {
        Q a = makeEdge(s[0], s[1]), b = makeEdge(s[1], s.back());
        if (sz(s) == 2) return { a, a->r() };
        splice(a->r(), b);
        auto side = s[0].cross(s[1], s[2]);
        Q c = side ? connect(b, a) : 0;
        return {side < 0 ? c->r() : a, side < 0 ? c : b->r() };
    } // c9e598
#define H(e) e->F(), e->p
#define valid(e) (e->F().cross(H(base)) > 0)
    Q A, B, ra, rb;
    int half = sz(s) / 2;
    tie(ra, A) = rec({all(s) - half});
    tie(B, rb) = rec({sz(s) - half + all(s)});
    while ((B->p.cross(H(A)) < 0 && (A = A->next())) ||
        (A->p.cross(H(B)) > 0 && (B = B->r()->o)));
    Q base = connect(B->r(), A);
    if (A->p == ra->p) ra = base->r();
    if (B->p == rb->p) rb = base;
#define DEL(e, init, dir) Q e = init->dir; if (valid(e)) \
    while (circ(e->dir->F(), H(base), e->F())) { \
        Q t = e->dir; \
        splice(e, e->prev()); \
        splice(e->r(), e->r()->prev()); \
        e->o = H; H = e; e = t; \
    } // a2e9b5
    for (;;) {
        DEL(LC, base->r(), o); DEL(RC, base, prev());
        if (!valid(LC) && !valid(RC)) break;
        if (!valid(LC) || (valid(RC) && circ(H(RC), H(LC))))
            base = connect(RC, base->r());
        else
            base = connect(base->r(), LC->r());
    } // fcf7ef
    return { ra, rb };
} // 7cf639
```

```
vector<P> triangulate(vector<P> pts) {
    sort(all(pts)); assert(unique(all(pts)) == pts.end());
    if (sz(pts) < 2) return {};
    Q e = rec(pts).first;
    vector<Q> q = {e};
    int qi = 0;
    while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
#define ADD { Q c = e; do { c->mark = 1; pts.push_back(c->p); \
    q.push_back(c->r()); c = c->next(); } while (c != e); } // 43
    e195 // 43e195
    ADD; pts.clear();
    while (qi < sz(q)) if (!(e = q[qi++])->mark) ADD;
    return pts;
} // a02307
```

4.5 3D

PolyhedronVolume.h

**Description:** Magic formula for the volume of a polyhedron. Faces should point outwards.

|                   |
|-------------------|
| 3058c3, 6L, 1m10s |
|-------------------|

|                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| template<class V, class L> double signedPolyVolume(const V& p, const L& trilst) {     double v = 0;     for (auto i : trilst) v += p[i.a].cross(p[i.b]).dot(p[i.c]);     return v / 6; } // fca9df |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Point3D.h

**Description:** Class to handle points in 3D space. T can be e.g. double or long long.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8058ae, 32L, 9m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| template<class T> struct Point3D {     typedef Point3D P;     typedef const P& R;     T x, y, z;     explicit Point3D(T x=0, T y=0, T z=0) : x(x), y(y), z(z) {}     bool operator<(R p) const {         return tie(x, y, z) < tie(p.x, p.y, p.z); } // 8eef6b     bool operator==(R p) const {         return tie(x, y, z) == tie(p.x, p.y, p.z); } // bd6a08     P operator+(R p) const { return P(x+p.x, y+p.y, z+p.z); }     P operator-(R p) const { return P(x-p.x, y-p.y, z-p.z); }     P operator*(T d) const { return P(x*d, y*d, z*d); }     P operator/(T d) const { return P(x/d, y/d, z/d); }     T dot(R p) const { return x*p.x + y*p.y + z*p.z; }     P cross(R p) const {         return P(y*p.z - z*p.y, z*p.x - x*p.z, x*p.y - y*p.x);     } // a77b7e     T dist2() const { return x*x + y*y + z*z; }     double dist() const { return sqrt((double)dist2()); }     //Azimuthal angle (longitude) to x-axis in interval [-pi, pi]     double phi() const { return atan2(y, x); }     //Zenith angle (latitude) to the z-axis in interval [0, pi]     double theta() const { return atan2(sqrt(x*x+y*y),z); }     P unit() const { return *this/(T)dist(); } //makes dist()==1     //returns unit vector normal to *this and p     P normal(P p) const { return cross(p).unit(); }     //returns point rotated 'angle' radians ccw around axis     P rotate(double angle, P axis) const {         double s = sin(angle), c = cos(angle); P u = axis.unit();         return u*dot(u)*(1-c) + (*this)*c - cross(u)*s;     } // 73af70 }; // 8058ae |

3dHull.h

**Description:** Computes all faces of the 3-dimension hull of a point set. \*No four points must be coplanar\*, or else random results will be returned. All faces will point outwards.

**Time:**  $\mathcal{O}(n^2)$

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5b45fc, 46L, 7m50s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| typedef Point3D<double> P3; struct PR {     void ins(int x) { (a == -1 ? a : b) = x; }     void rem(int x) { (a == x ? a : b) = -1; }     int cnt() { return (a != -1) + (b != -1); }     int a, b; }; // cf7c9e struct F { P3 q; int a, b, c; }; vector<F> hull3d(const vector<P3>& A) {     assert(sz(A) >= 4);     vector<vector<PR>> E(sz(A), vector<PR>(sz(A), {-1, -1})); #define E(x,y) E[f.x][f.y]     vector<F> FS;     auto mf = [&](int i, int j, int k, int l) {         P3 q = (A[j] - A[i]).cross((A[k] - A[i]));         if (q.dot(A[l]) > q.dot(A[i]))             q = q * -1; |

```
F f{q, i, j, k};
E(a,b).ins(k); E(a,c).ins(j); E(b,c).ins(i);
FS.push_back(f);
}; // d73a06
rep(i,0,4) rep(j,i+1,4) rep(k,j+1,4)
mf(i, j, k, 6 - i - j - k);

rep(i,4,sz(A)) {
rep(j,0,sz(FS)) {
F f = FS[j];
if(f.q.dot(A[i]) > f.q.dot(A[f.a])) {
E(a,b).rem(f.c);
E(a,c).rem(f.b);
E(b,c).rem(f.a);
swap(FS[j--], FS.back());
FS.pop_back();
} // 5cd5dc
} // 220067
int nw = sz(FS);
rep(j,0,nw) {
F f = FS[j];
#define C(a, b, c) if (E(a,b).cnt() != 2) mf(f.a, f.b, i, f.c);
C(a, b, c); C(a, c, b); C(b, c, a);
} // 248ed4
} // 47289c
for (F& it : FS) if ((A[it.b] - A[it.a]).cross(
A[it.c] - A[it.a]).dot(it.q) <= 0) swap(it.c, it.b);
return FS;
}; // be2ca2
```

sphericalDistance.h

Description: Returns the shortest distance on the sphere with radius radius between the points with azimuthal angles (longitude) f1 ( $\phi_1$ ) and f2 ( $\phi_2$ ) from x axis and zenith angles (colatitude) t1 ( $\theta_1$ ) and t2 ( $\theta_2$ ) from z axis (0 = north pole). All angles measured in radians. The algorithm starts by converting the spherical coordinates to cartesian coordinates so if that is what you have you can use only the two last rows. dx\*radius is then the difference between the two points in the x direction and d\*radius is the total distance between the points.

80b54c, 8L, 2m

```
double sphericalDistance(double f1, double t1,
double f2, double t2, double radius) {
double dx = sin(t2)*cos(f2) - sin(t1)*cos(f1);
double dy = sin(t2)*sin(f2) - sin(t1)*sin(f1);
double dz = cos(t2) - cos(t1);
double d = sqrt(dx*dx + dy*dy + dz*dz);
return radius*2*asin(min(1.,d/2));
} // cb1a4b
```

## Graph (5)

### 5.1 Fundamentals

BellmanFord.h

Description: Calculates shortest paths from s in a graph that might have negative edge weights. Unreachable nodes get dist = inf; nodes reachable through negative-weight cycles get dist = -inf. Assumes  $V^2 \max |w_i| < \sim 2^{83}$ . Time:  $\mathcal{O}(VE)$

830a8f, 21L, 4m30s

```
const ll inf = LLONG_MAX;
struct Ed { int a, b, w, s() { return a < b ? a : -a; }};
struct Node { ll dist = inf; int prev = -1; };
void bellmanFord(vector<Node>& nodes, vector<Ed>& eds, int s) {
nodes[s].dist = 0;
sort(all(eds), [](Ed a, Ed b) { return a.s() < b.s(); });
int lim = sz(nodes) / 2 + 2; // /3+100 with shuffled vertices
rep(i,0,lim) for (Ed ed : eds) {
Node cur = nodes[ed.a], &dest = nodes[ed.b];
```

```
if (abs(cur.dist) == inf) continue;
ll d = cur.dist + ed.w;
if (d < dest.dist) {
dest.prev = ed.a;
dest.dist = (i < lim-1 ? d : -inf);
} // 452019
} // 75a370
rep(i,0,lim) for (Ed e : eds) {
if (nodes[e.a].dist == -inf)
nodes[e.b].dist = -inf;
} // 1d7315
} // fa39de
```

FunctGraph.h

Description: Functional Graph

Memory:  $\mathcal{O}(n)$

Time:  $\mathcal{O}(n)$

152fc5, 24L, 4m

```
struct FunctGraph{
int n;
vi head, comp;
vector<vi> gr, cycles;
FunctGraph(vi& fn):
n(sz(fn)), head(n, -1), comp(n), gr(n) {
rep(i, 0, n)gr[fn[i]].pb(i);
vi visited(n, 0);
auto dfs = [&](auto rec, int v, int c) -> void {
head[v] = c; visited[v] = 1;
for(int f : gr[v])if (head[f]!=f)rec(rec, f, c);
}; // e1fa06
rep(i, 0, n){
if (visited[i])continue;
int l=fn[i], r=fn[fn[i]];
while(l!=r) l=fn[l], r=fn[fn[r]];
vi cur = {r};
for(l=fn[l]; l!=r; l=fn[l]) cur.pb(l);
for(int x : cur) head[x] = x, comp[x] = sz(cycles);
cycles.pb(cur);
for(int x : cur) dfs(dfs, x, x);
} // 01a153
} // 0a2937
}; // 152fc5
```

### 5.2 Flow

Dinic.h

Description: Flow algorithm with complexity  $\mathcal{O}(VE \log U)$  where  $U = \max |cap|$ .  $\mathcal{O}(\min(E^{1/2}, V^{2/3})E)$  if  $U = 1$ ;  $\mathcal{O}(\sqrt{VE})$  for bipartite matching.

d7f0f1, 42L, 7m30s

```
struct Dinic {
struct Edge {
int to, rev;
ll c, oc;
ll flow() { return max(oc - c, 0LL); } // if you need flows
}; // d46531
vi lvl, ptr, q;
vector<vector<Edge>> adj;
Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {}
void addEdge(int a, int b, ll c, ll rcap = 0) {
adj[a].push_back({b, sz(adj[b]), c, c});
adj[b].push_back({a, sz(adj[a]) - 1, rcap, rcap});
} // 01ad91
ll dfs(int v, int t, ll f) {
if (v == t || !f) return f;
for (int& i = ptr[v]; i < sz(adj[v]); i++) {
Edge& e = adj[v][i];
if (lvl[e.to] == lvl[v] + 1)
if (ll p = dfs(e.to, t, min(f, e.c))) {
```

```
e.c -= p, adj[e.to][e.rev].c += p;
return p;
} // 494522
} // f73941
return 0;
} // 0922b8
ll calc(int s, int t) {
ll flow = 0; q[0] = s;
rep(L,0,31) do { // 'int L=30' maybe faster for random data
lvl = ptr = vi(sz(q));
int qi = 0, qe = lvl[s] = 1;
while (qi < qe && !lvl[t]) {
int v = q[qi++];
for (Edge e : adj[v])
if (!lvl[e.to] && e.c >> (30 - L))
q[qe++] = e.to, lvl[e.to] = lvl[v] + 1;
} // 80ef52
while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
} while (lvl[t]); // 96458d
return flow;
} // 3eb6d1
bool leftOfMinCut(int a) { return lvl[a] != 0; }
}; // d7f0f1
```

MinCostMaxFlow.h

Description: Min-cost max-flow. If costs can be negative, call setpi before maxflow, but note that negative cost cycles are not supported. To obtain the actual flow, look at positive values only.

Time:  $\mathcal{O}(FE \log(V))$  where F is max flow.  $\mathcal{O}(VE)$  for setpi.

<bits/extc++.h>

135b73, 69L, 13m

```
const ll INF = numeric_limits<ll>::max() / 4;
struct MCMF {
struct edge {
int from, to, rev;
ll cap, cost, flow;
}; // 092ff8
int N;
vector<vector<edge>> ed;
vi seen;
vector<ll> dist, pi;
vector<edge*> par;
MCMF(int N) : N(N), ed(N), seen(N), dist(N), pi(N), par(N) {}
void addEdge(int from, int to, ll cap, ll cost) {
if (from == to) return;
ed[from].push_back(edge{ from,to,sz(ed[to]),cap,cost,0 });
ed[to].push_back(edge{ to,from,sz(ed[from])-1,0,-cost,0 });
} // c71528
void path(int s) {
fill(all(seen), 0);
fill(all(dist), INF);
dist[s] = 0; ll di;
__gnu_pbds::priority_queue<pair<ll, int>> q;
vector<decltype(q)::point_iterator> its(N);
q.push({ 0, s });
while (!q.empty()) {
s = q.top().second; q.pop();
seen[s] = 1; di = dist[s] + pi[s];
for (edge& e : ed[s]) if (!seen[e.to]) {
ll val = di - pi[e.to] + e.cost;
if (e.cap - e.flow > 0 && val < dist[e.to]) {
dist[e.to] = val;
par[e.to] = &e;
if (its[e.to] == q.end())
its[e.to] = q.push({ -dist[e.to], e.to });
else
q.modify(its[e.to], { -dist[e.to], e.to });
} // ca07f4
} // 4cd18f
```

```
    } // 062b8f
    rep(i,0,N) pi[i] = min(pi[i] + dist[i], INF);
} // 7e4cbe
pair<ll, ll> maxflow(int s, int t) {
    ll totflow = 0, totcost = 0;
    while (path(s), seen[t]) {
        ll fl = INF;
        for (edge* x = par[t]; x; x = par[x->from])
            fl = min(fl, x->cap - x->flow);

        totflow += fl;
        for (edge* x = par[t]; x; x = par[x->from]) {
            x->flow += fl;
            ed[x->to][x->rev].flow -= fl;
        } // 3bfaf3
    } // 8d9a6a
    rep(i,0,N) for(edge& e : ed[i]) totcost += e.cost * e.flow;
    return {totflow, totcost/2};
} // 24f5a0
// If some costs can be negative, call this before maxflow:
void setpi(int s) { // (otherwise, leave this out)
    fill(all(pi), INF); pi[s] = 0;
    int it = N, ch = 1; ll v;
    while (ch-- && it--)
        rep(i,0,N) if (pi[i] != INF)
            for (edge& e : ed[i]) if (e.cap)
                if ((v = pi[i] + e.cost) < pi[e.to])
                    pi[e.to] = v, ch = 1;
    assert(it >= 0); // negative cost cycle
} // 6847d8
}; // b3692f
```

GlobalMinCut.h

Description: Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.

Time:  $\mathcal{O}(V^3)$

8b0e19, 21L, 3m50s

```
pair<int, vi> globalMinCut(vector<vi> mat) {
    pair<int, vi> best = {INT_MAX, {}};
    int n = sz(mat);
    vector<vi> co(n);
    rep(i,0,n) co[i] = {i};
    rep(ph,1,n) {
        vi w = mat[0];
        size_t s = 0, t = 0;
        rep(it,0,n-ph) { //  $\mathcal{O}(V^2) \rightarrow \mathcal{O}(E \log V)$  with prio. queue
            w[t] = INT_MIN;
            s = t, t = max_element(all(w)) - w.begin();
            rep(i,0,n) w[i] += mat[t][i];
        } // ec93df
        best = min(best, {w[t] - mat[t][t], co[t]});
        co[s].insert(co[s].end(), all(co[t]));
        rep(i,0,n) mat[s][i] += mat[t][i];
        rep(i,0,n) mat[i][s] = mat[s][i];
        mat[0][t] = INT_MIN;
    } // ca0062
    return best;
} // 8b0e19
```

GomoryHu.h

Description: Given a list of edges representing an undirected flow graph, returns edges of the Gomory-Hu tree. The max flow between any pair of vertices is given by minimum edge weight along the Gomory-Hu tree path.

Time:  $\mathcal{O}(V)$  Flow Computations

"Dinic.h" e2b333, 13L, 2m20s

```
typedef array<ll, 3> Edge;
vector<Edge> gomoryHu(int N, vector<Edge> ed) {
    vector<Edge> tree;
```

```
    vi par(N);
    rep(i,1,N) {
        Dinic D(N); // Dinic also works
        for (Edge t : ed) D.addEdge(t[0], t[1], t[2], t[2]);
        tree.push_back({i, par[i], D.calc(i, par[i])});
        rep(j,i+1,N)
            if (par[j] == par[i] && D.leftOfMinCut(j)) par[j] = i;
    } // 42d832
    return tree;
} // 4edcec
```

5.3 Matching

HopcroftKarp.h

Description: Fast bipartite matching algorithm. Graph  $g$  should be a list of neighbors of the left partition, and  $r$  should be a vector full of  $-1$ 's of the same size as the right partition. Returns the size of the matching.  $r[i]$  will be the match for vertex  $i$  on the right side, or  $-1$  if it's not matched.

Time:  $\mathcal{O}(E\sqrt{V})$

731cfb, 20L, 4m

```
int hopcroftKarp(vector<vi>& g, vi& r) {
    int n = sz(g), res = 0;
    vi l(n, -1), q(n), d(n);
    auto dfs = [&](auto f, int u) -> bool {
        int t = exchange(d[u], 0) + 1;
        for (int v : g[u])
            if (r[v] == -1 || (d[r[v]] == t && f(f, r[v])))
                return l[u] = v, r[v] = u, 1;
        return 0;
    }; // a95e38
    for (int t = 0, f = 0;; t = f = 0, d.assign(n, 0)) {
        rep(i,0,n) if (l[i] == -1) q[t++] = i, d[i] = 1;
        rep(i,0,t) for (int v : g[q[i]]) {
            if (r[v] == -1) f = 1;
            else if (!d[r[v]]) d[r[v]] = d[q[i]] + 1, q[t++] = r[v];
        } // 64af74
        if (!f) return res;
        rep(i,0,n) if (l[i] == -1) res += dfs(dfs, i);
    } // cdf3b2
} // 731cfb
```

MinimumVertexCover.h

Description: Finds a minimum vertex cover in a bipartite graph. The size is the same as the size of a maximum matching, and the complement is a maximum independent set.

"HopcroftKarp.h" bfb654, 20L, 3m50s

```
vi cover(vector<vi>& g, int n, int m) {
    vi match(m, -1);
    int res = hopcroftKarp(g, match);
    vector<bool> lfound(n, true), seen(m);
    for (int it : match) if (it != -1) lfound[it] = false;
    vi q, cover;
    rep(i,0,n) if (lfound[i]) q.push_back(i);
    while (!q.empty()) {
        int i = q.back(); q.pop_back();
        lfound[i] = 1;
        for (int e : g[i]) if (!seen[e] && match[e] != -1) {
            seen[e] = true;
            q.push_back(match[e]);
        } // 46e035
    } // 069994
    rep(i,0,n) if (!lfound[i]) cover.push_back(i);
    rep(i,0,m) if (seen[i]) cover.push_back(n+i);
    assert(sz(cover) == res);
    return cover;
} // bfb654
```

GeneralMatching.h

Description: Matching for general graphs. 1-indexed vertices.

Memory:  $\mathcal{O}(n + m)$

Time:  $\mathcal{O}(\sqrt{nm} \log_{\max\{2,1+m/n\}} n)$

9437b5, 313L, 75m

```
class MaximumMatching {
public:
    struct Edge {int from, to;};
    static constexpr int Inf = 1 << 30;
private:
    enum Label {
        kInner = -1, // should be < 0
        kFree = 0 // should be 0
    }; // c09e7e
    struct Link {int from, to;};
    struct Log {int v, par;};
    struct LinkedList {
        LinkedList() {}
        LinkedList(int N, int M) : N(N), next(M) { clear(); }
        void clear() { head.assign(N, -1); }
        void push(int h, int u) { next[u] = head[h], head[h] = u; }
        int N; vector<int> head, next;
    }; // 1a51c7
    template <typename T> struct Queue {
        Queue() {}
        Queue(int N) : qh(0), qt(0), data(N) {}
        T operator[](int i) const { return data[i]; }
        void enqueue(int u) { data[qt++] = u; }
        int dequeue() { return data[qh++]; }
        bool empty() const { return qh == qt; }
        void clear() { qh = qt = 0; }
        int size() const { return qt; }
        int qh, qt; vector<T> data;
    }; // 4bfceb
    struct DisjointSetUnion {
        DisjointSetUnion() {}
        DisjointSetUnion(int N) : par(N) { for (int i = 0; i < N; ++i) par[i] = i; }
        int find(int u) { return par[u] == u ? u : (par[u] = find(par[u])); }
        void unite(int u, int v) {
            u = find(u), v = find(v);
            if (u != v) par[v] = u;
        } // 461e55
        vector<int> par;
    }; // a21f68
public:
    MaximumMatching(int N, const vector<Edge> &in)
        : N(N), NH(N >> 1), ofs(N + 2, 0), edges(in.size() * 2) {}
    for (auto &e : in) ofs[e.from + 1] += 1, ofs[e.to + 1] += 1;
    for (int i = 1; i <= N + 1; ++i) ofs[i] += ofs[i - 1];
    for (auto &e : in) {
        edges[ofs[e.from]++] = e;
        edges[ofs[e.to]++] = {e.to, e.from};
    } // 36bba8
    for (int i = N + 1; i > 0; --i) ofs[i] = ofs[i - 1];
    ofs[0] = 0;
} // a78156
int maximum_matching() {
    initialize(); int match = 0;
    while (match * 2 + 1 < N) {
        reset_count();
        bool has_augmenting_path = do_edmonds_search();
        if (!has_augmenting_path) break;
        match += find_maximal();
        clear();
    }
}
```

```

    } // 8c0a2c
    return match;
} // da7b99
vector<Edge> get_edges() {
    vector<Edge> ans;
    for (int i = 1; i <= N; ++i) if (mate[i] > i) ans.
        push_back(Edge{i, mate[i]});
    return ans;
} // f8586b
private:
void reset_count() {
    time_current_ = 0;
    time_augment_ = Inf;
    contract_count_ = 0;
    outer_id_ = 1;
    dsu_changelog_size_ = dsu_changelog_last_ = 0;
} // 6c73d5
void clear() {
    que.clear();
    for (int u = 1; u <= N; ++u) potential[u] = 1;
    for (int u = 1; u <= N; ++u) dsu.par[u] = u;
    for (int t = time_current_; t <= N / 2; ++t) list.head[
        t] = -1;
    for (int u = 1; u <= N; ++u) blossom.head[u] = -1;
} // 6e020f
inline void grow(int x, int y, int z) {
    label[y] = kInner;
    potential[y] = time_current_; // visited time
    link[z] = {x, y};
    label[z] = label[x];
    potential[z] = time_current_ + 1;
    que.enqueue(z);
} // 5a596b
void contract(int x, int y) {
    int bx = dsu.find(x), by = dsu.find(y);
    const int h = -(++contract_count_) + kInner;
    label[mate[bx]] = label[mate[by]] = h;
    int lca = -1;
    while (1) {
        if (mate[by] != 0) swap(bx, by);
        bx = lca = dsu.find(link[bx].from);
        if (label[mate[bx]] == h) break;
        label[mate[bx]] = h;
    } // d03cf2
    for (auto bv : {dsu.par[x], dsu.par[y]}) {
        for (; bv != lca; bv = dsu.par[link[bv].from]) {
            int mv = mate[bv];
            link[mv] = {x, y};
            label[mv] = label[x];
            potential[mv] =
                1 + (time_current_ - potential[mv]) +
                time_current_;
            que.enqueue(mv);
            dsu.par[bv] = dsu.par[mv] = lca;
            dsu_changelog[dsu_changelog_last_++] = {bv, lca
            };
            dsu_changelog[dsu_changelog_last_++] = {mv, lca
            };
        } // 8fef02
    } // 2b0c42
} // a9547e
bool find_augmenting_path() {
    while (!que.empty()) {
        int x = que.dequeue(), lx = label[x], px =
            potential[x],
            bx = dsu.find(x);
        for (int eid = ofs[x]; eid < ofs[x + 1]; ++eid) {
            int y = edges[eid].to;
            if (label[y] > 0) { // outer blossom/vertex

```

```

                int time_next = (px + potential[y]) >> 1;
                if (lx != label[y]) {
                    if (time_next == time_current_) return
                        true;
                    time_augment_ = min(time_next,
                        time_augment_);
                } else { // a6d4c5
                    if (bx == dsu.find(y)) continue;
                    if (time_next == time_current_) {
                        contract(x, y);
                        bx = dsu.find(x);
                    } else if (time_next <= NH) // 1efde3
                        list.push(time_next, eid);
                } // 1666ad
            } else if (label[y] == kFree) { // free vertex
                // c17e44
                int time_next = px + 1;
                if (time_next == time_current_)
                    grow(x, y, mate[y]);
                else if (time_next <= NH)
                    list.push(time_next, eid);
            } // 48780d
        } // 30c821
    } // 1f7dd0
    return false;
} // 5f1707
bool adjust_dual_variables() {
    const int time_lim = min(NH + 1, time_augment_);
    for (++time_current_; time_current_ <= time_lim; ++
        time_current_) {
        dsu_changelog_size_ = dsu_changelog_last_;
        if (time_current_ == time_lim) break;
        bool updated = false;
        for (int h = list.head[time_current_]; h >= 0; h =
            list.next[h]) {
            auto &e = edges[h];
            int x = e.from, y = e.to;
            if (label[y] > 0) {
                if (potential[x] + potential[y] != (
                    time_current_ << 1))
                    continue;
                if (dsu.find(x) == dsu.find(y)) continue;
                if (label[x] != label[y]) {
                    time_augment_ = time_current_;
                    return false;
                } // e7c864
                contract(x, y);
                updated = true;
            } else if (label[y] == kFree) { // 1a7dca
                grow(x, y, mate[y]);
                updated = true;
            } // ebac0d
        } // 7bf910
        list.head[time_current_] = -1;
        if (updated) return false;
    } // a2963f
    return time_current_ > NH;
} // 361891
bool do_edmonds_search() {
    label[0] = kFree;
    for (int u = 1; u <= N; ++u) {
        if (mate[u] == 0) {
            que.enqueue(u);
            label[u] = u; // component id
        } else // 52e61c
            label[u] = kFree;
    } // 585bc4
    while (1) {
        if (find_augmenting_path()) break;

```

```

        bool maximum = adjust_dual_variables();
        if (maximum) return false;
        if (time_current_ == time_augment_) break;
    } // 3b82c1
    for (int u = 1; u <= N; ++u) {
        if (label[u] > 0)
            potential[u] -= time_current_;
        else if (label[u] < 0)
            potential[u] = 1 + (time_current_ - potential[u
            ]);
    } // c77cf7
    return true;
} // e1fc96
void rematch(int v, int w) {
    int t = mate[v];
    mate[v] = w;
    if (mate[t] != v) return;
    if (link[v].to == dsu.find(link[v].to)) {
        mate[t] = link[v].from;
        rematch(mate[t], t);
    } else { // 45eff2
        int x = link[v].from, y = link[v].to;
        rematch(x, y);
        rematch(y, x);
    } // 4a404f
} // c4e2e9
bool dfs_augment(int x, int bx) {
    int px = potential[x], lx = label[bx];
    for (int eid = ofs[x]; eid < ofs[x + 1]; ++eid) {
        int y = edges[eid].to;
        if (px + potential[y] != 0) continue;
        int by = dsu.find(y), ly = label[by];
        if (ly > 0) { // outer
            if (lx >= ly) continue;
            int stack_beg = stack_last_;
            for (int bv = by; bv != bx; bv = dsu.find(link[
                bv].from)) {
                int bw = dsu.find(mate[bv]);
                stack[stack_last_++] = bw;
                link[bw] = {x, y};
                dsu.par[bv] = dsu.par[bw] = bx;
            } // 211d3e
            while (stack_last_ > stack_beg) {
                int bv = stack[--stack_last_];
                for (int v = blossom.head[bv]; v >= 0;
                    v = blossom.next[v]) {
                    if (!dfs_augment(v, bx)) continue;
                    stack_last_ = stack_beg;
                    return true;
                } // 9ded5e
            } // 476c04
        } else if (ly == kFree) { // e5357f
            label[by] = kInner;
            int z = mate[by];
            if (z == 0) {
                rematch(x, y);
                rematch(y, x);
                return true;
            } // 781371
            int bz = dsu.find(z);
            link[bz] = {x, y};
            label[bz] = outer_id_++;
            for (int v = blossom.head[bz]; v >= 0; v =
                blossom.next[v]) {
                if (dfs_augment(v, bz)) return true;
            } // 5ea9e4
        } // 1c508f
    } // 7f2ab3
    return false;

```

```

} // 44c41a
int find_maximal() {
    for (int u = 1; u <= N; ++u) dsu.par[u] = u;
    for (int i = 0; i < dsu_changelog_size_; ++i) {
        dsu.par[dsu_changelog[i].v] = dsu_changelog[i].par;
    } // 384510
    for (int u = 1; u <= N; ++u) {
        label[u] = kFree;
        blossom.push(dsu.find(u), u);
    } // 2a02fd
    int ret = 0;
    for (int u = 1; u <= N; ++u)
        if (!mate[u]) {
            int bu = dsu.par[u];
            if (label[bu] != kFree) continue;
            label[bu] = outer_id++;
            for (int v = blossom.head[bu]; v >= 0; v = blossom.next[v]) {
                if (!dfs_augment(v, bu)) continue;
                ret += 1;
                break;
            } // e4fb6a
        } // 4db9e3
    assert(ret >= 1); return ret;
} // 500171
void initialize() {
    que = Queue<int>(N);
    mate.assign(N + 1, 0);
    potential.assign(N + 1, 1);
    label.assign(N + 1, kFree);
    link.assign(N + 1, {0, 0});
    dsu_changelog.resize(N);
    dsu = DisjointSetUnion(N + 1);
    list = LinkedList(NH + 1, edges.size());
    blossom = LinkedList(N + 1, N + 1);
    stack.resize(N);
    stack_last_ = 0;
} // be011c
private:
const int N, NH;
vector<int> ofs;
vector<Edge> edges;
Queue<int> que;
vector<int> mate, potential;
vector<int> label;
vector<Link> link;
vector<Log> dsu_changelog;
int dsu_changelog_last_, dsu_changelog_size_;
DisjointSetUnion dsu;
LinkedList list, blossom;
vector<int> stack;
int stack_last_;
int time_current_, time_augment_;
int contract_count_, outer_id_;
}; // b7724d
using Edge = MaximumMatching::Edge;
void example() { // Graph of Love problem
    int n; cin >> n;
    vector<Edge> edges(n, {0, 0});
    for (int i = 1, j; i <= n; i++) cin >> j, edges[i-1] = {i, j};
    auto M = MaximumMatching(n, edges);
    cout << M.maximum_matching() << '\n';
} // 0523ec
```

OnlineMatching.h

Description: Modified khun developed for specific question able to run 2\*10<sup>6</sup> queries, in 2\*10<sup>6</sup> x 10<sup>6</sup> graph in 3 seconds codeforces

Time:  $\mathcal{O}(\textit{confia})$

6ac539, 36L, 5m20s

```

struct OnlineMatching {
    int n = 0, m = 0;
    vector<int> vis, match, dist;
    vector<vector<int>> g;
    vector<int> last;
    int t = 0;

    OnlineMatching(int n_, int m_) : n(n_), m(m_), vis(n, 0),
        match(m, -1), dist(n, n+1), g(n), last(n, -1) {}
    void add(int a, int b) { g[a].pb(b); }
    bool kuhn(int a) {
        vis[a] = t;
        for(int b: g[a]) {
            int c = match[b];
            if (c == -1) {
                match[b] = a;
                return true;
            } // 38b210
            if (last[c] != t || (dist[a] + 1 < dist[c]))
                dist[c] = dist[a] + 1, last[c] = t;
        } // d30675
        for (int b: g[a]) {
            int c = match[b];
            if (dist[a] + 1 == dist[c] && vis[c] != t && kuhn(c)) {
                match[b] = a;
                return true;
            } // 2dac75
        } // e58bd5
        return false;
    } // b533ee
    bool can_match(int a) {
        t++;
        last[a] = t;
        dist[a] = 0;
        return kuhn(a);
    } // 32302b
}; // 6ac539
```

5.4 DFS algorithms

SCC.h

Description: Finds strongly connected components in a directed graph. If vertices  $u, v$  belong to the same component, we can reach  $u$  from  $v$  and vice versa.

Usage: scc(graph, [&](vi& v) { ... }) visits all components in reverse topological order. comp[i] holds the component index of a node (a component only has edges to components with lower index). ncomps will contain the number of components.

Time:  $\mathcal{O}(E + V)$

76b5c9, 24L, 3m50s

```

vi val, comp, z, cont;
int Time, ncomps;
template<class G, class F> int dfs(int j, G& g, F& f) {
    int low = val[j] = ++Time, x; z.push_back(j);
    for (auto e : g[j]) if (comp[e] < 0)
        low = min(low, val[e] ?: dfs(e,g,f));

    if (low == val[j]) {
        do {
            x = z.back(); z.pop_back();
            comp[x] = ncomps;
            cont.push_back(x);
        } while (x != j); // ae85bd
        f(cont); cont.clear();
        ncomps++;
    } // 64c1b9
```

```

    return val[j] = low;
} // 3513bd
template<class G, class F> void scc(G& g, F f) {
    int n = sz(g);
    val.assign(n, 0); comp.assign(n, -1);
    Time = ncomps = 0;
    rep(i,0,n) if (comp[i] < 0) dfs(i, g, f);
} // 56b050
```

BiconnectedComponents.h

Description: Finds all biconnected components in an undirected graph. In a biconnected component there are at least two internally disjoint paths between any two nodes a cycle exists through them. Note that a node can be in several components. Every edge is in a single component. Nodes without edges will not be in any components

Time:  $\mathcal{O}(E + V)$

61b438, 50L, 8m

```

vector<pair<int, int>> edges; // edges
vector<vector<pair<int, int>>> g; // [b, edge idx]
vi tin, st, art;
int dfstime = 0;
vector<vi> bcc;
int dfs(int a, int p) {
    int top = tin[a] = ++dfstime;
    bool child = (p != -1);
    for (auto [b, e]: g[a]) {
        if (tin[b]) {
            top = min(top, tin[b]);
            if (tin[b] < tin[a]) st.pb(e);
        } // 0df373
        else {
            int si = sz(st);
            int up = dfs(b, e);
            top = min(top, up);
            if (up > tin[a]) { /*e is a bridge */
                if (up >= tin[a]) {
                    st.pb(e);
                    bcc.eb(st.begin() + si, st.end());
                    st.resize(si);
                    art[a] += child;
                    child = true;
                } // 46103a
                else if (up < tin[a]) st.pb(e);
            } // 38ea8a
        } // 53a60e
    }
    return top;
} // 17df2f
void bicomps() {
    int n = sz(g);
    tin.assign(n, 0), art.assign(n, 0);
    rep(i,0,n) if (!tin[i]) dfs(i, -1);
} // 600614
vi comp;
vector<vi> tree;
void build_tree() { // Optional
    int n = sz(g);
    comp.resize(n), tree.resize(n + sz(bcc));
    rep(i, 0, sz(bcc)) {
        for (int eid: bcc[i]) {
            auto [a, b] = edges[eid];
            if (art[a] && (empty(tree[a]) || tree[a].back() != n+i))
                tree[a].pb(n+i), tree[n+i].pb(a);
            if (art[b] && (empty(tree[b]) || tree[b].back() != n+i))
                tree[b].pb(n+i), tree[n+i].pb(b);
            comp[a] = comp[b] = n + i;
        } // 2e7ad5
    } // 8e4c66
    rep(i, 0, n) if (art[i]) comp[i] = i;
} // b645e9
```

**TwoCC.h**  
**Description:** Finds all two edge connected components in an undirected graph.  
**Time:**  $\mathcal{O}(E + V)$

af5478, 41L, 6m

```
vector<vector<pair<int, int>>> g;
vector<pair<int, int>> edges;
vi tin, st, bridges;
int dfstime = 0;
vector<vi> twocc;
int dfs(int a, int p) {
    int top = tin[a] = ++dfstime;
    int si = st.size();
    st.pb(a);
    for (auto [b, e]: g[a]) if (e != p) {
        if (tin[b]) top = min(top, tin[b]);
        else {
            int up = dfs(b, e);
            top = min(top, up);
            if (up > tin[a]) bridges.pb(e);
        } // 1dd281
    } // ada038
    if (top == tin[a]) {
        twocc.eb(st.begin() + si, st.end());
        st.resize(si);
    } // 0f03f3
    return top;
} // 55bb17
void twocomps() {
    int n = sz(g);
    tin.assign(n, 0);
    rep(i, 0, n) if (!tin[i]) dfs(i, -1);
} // 9a2d22
vi comp;
vector<vi> tree;
void build_tree() { // Optional
    int n = sz(g);
    tree.resize(sz(twocc)); comp.resize(n);
    rep(i, 0, sz(twocc))
        for (int a: twocc[i]) comp[a] = i;

    for (int eid: bridges) {
        auto [a, b] = edges[eid];
        tree[comp[a]].pb(comp[b]), tree[comp[b]].pb(comp[a]);
    } // 1a7e1f
} // c306f0
```

**2sat.h**  
**Description:** Calculates a valid assignment to boolean variables a, b, c,... to a 2-SAT problem, so that an expression of the type  $(a||b)\&\&(!a||c)\&\&(d||!b)\&\&...$  becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions (~x).  
**Usage:** TwoSat ts(number of boolean variables);  
ts.either(0, ~3); // Var 0 is true or var 3 is false  
ts.setValue(2); // Var 2 is true  
ts.atMostOne({0,~1,2}); // <= 1 of vars 0, ~1 and 2 are true  
ts.solve(); // Returns true iff it is solvable  
ts.values[0..N-1] holds the assigned values to the vars  
**Time:**  $\mathcal{O}(N + E)$ , where N is the number of boolean variables, and E is the number of clauses.

8e1208, 48L, 7m40s

```
struct TwoSat {
    int N; vector<vi> gr; vi values; // 0 = false, 1 = true
    TwoSat(int n = 0) : N(n), gr(2*n) {}
    void either(int f, int j) {
        f = max(2*f, -1-2*f);
        j = max(2*j, -1-2*j);
        gr[f].push_back(j^1);
        gr[j].push_back(f^1);
    }
```

```
} // 516db0
void setValue(int x) { either(x, x); }
int addVar() { // (optional)
    gr.emplace_back();
    gr.emplace_back();
    return N++;
} // 7b5f84
void atMostOne(const vi& li) { // (optional)
    if (sz(li) <= 1) return;
    int cur = ~li[0];
    rep(i,2,sz(li)) {
        int next = addVar();
        either(cur, ~li[i]);
        either(cur, next);
        either(~li[i], next);
        cur = ~next;
    } // 8d3782
    either(cur, ~li[1]);
} // 10f2ea
vi val, comp, z; int time = 0;
int dfs(int i) {
    int low = val[i] = ++time, x; z.push_back(i);
    for(int e : gr[i]) if (!comp[e])
        low = min(low, val[e] ?: dfs(e));
    if (low == val[i]) do {
        x = z.back(); z.pop_back();
        comp[x] = low;
        if (values[x]>>1) == -1)
            values[x]>>1) = x&1;
    } while (x != i); // b15351
    return val[i] = low;
} // ef583a
bool solve() {
    values.assign(N, -1);
    val.assign(2*N, 0); comp = val;
    rep(i,0,2*N) if (!comp[i]) dfs(i);
    rep(i,0,N) if (comp[2*i] == comp[2*i+1]) return 0;
    return 1;
} // 2bb76d
}; // 8e1208
```

**EulerWalk.h**  
**Description:** Eulerian undirected/directed path/cycle algorithm. Input should be a vector of (dest, global edge index), where for undirected graphs, forward/backward edges have the same index. Returns a list of nodes in the Eulerian path/cycle with src at both start and end, or empty list if no cycle/path exists. To get edge indices back, add .second to s and ret.  
**Time:**  $\mathcal{O}(V + E)$

780b64, 15L, 3m30s

```
vi eulerWalk(vector<vector<pii>>& gr, int nedges, int src=0) {
    int n = sz(gr);
    vi D(n), its(n), eu(nedges), ret, s = {src};
    D[src]++; // to allow Euler paths, not just cycles
    while (!s.empty()) {
        int x = s.back(), y, e, &it = its[x], end = sz(gr[x]);
        if (it == end){ ret.push_back(x); s.pop_back(); continue; }
        tie(y, e) = gr[x][it++];
        if (!eu[e]) {
            D[x]--, D[y]++;
            eu[e] = 1; s.push_back(y);
        } // 22a87a // 94de26
    }
    for (int x : D) if (x < 0 || sz(ret) != nedges+1) return {};
    return {ret.rbegin(), ret.rend()};
} // 780b64
```

**DominatorTree.h**  
**Description:** Dominator Tree, creates the graph tree, where all ancestors of a u in the tree are necessary in the path from the root to u  
**Memory:**  $\mathcal{O}(n)$   
**Time:**  $\mathcal{O}((n + m)\log(n))$  build

d7c79f, 54L, 9m10s

```
struct DominatorTree {
    int n, dfstime;
    vector<vector<int>> g, gt, tree, bucket, down;
    vector<int> S, dsu, label, sdom, idom, id;

    DominatorTree(vector<vector<int>> &_g, int root)
        : n(sz(_g)), dfstime(0), g(_g), gt(n), tree(n), bucket(n),
          down(n),
          S(n), dsu(n), label(n), sdom(n), idom(n), id(n) {
        prep(root); reverse(S.begin(), S.begin() + dfstime);
        for(int u : S) {
            for(int v : gt[u]) {
                int w = fnd(v);
                if(id[ sdom[w] ] < id[ sdom[u] ])
                    sdom[u] = sdom[w];
            } // e059b2
            gt[u].clear();
            if(u != root) bucket[ sdom[u] ].push_back(u);
            for(int v : bucket[u]) {
                int w = fnd(v);
                if(sdom[w] == sdom[v]) idom[v] = sdom[v];
                else idom[v] = w;
            } // 72077b
            bucket[u].clear();
            for(int v : down[u]) dsu[v] = u;
            down[u].clear();
        } // 3197c4
        reverse(S.begin(), S.begin() + dfstime);
        for(int u : S) if(u != root) {
            if(idom[u] != sdom[u]) idom[u] = idom[ idom[u] ];
            tree[ idom[u] ].push_back(u);
        } // 96e582
        idom[root] = root;
    } // b239ba
    void prep(int u){
        S[dfstime] = u;
        id[u] = ++dfstime;
        label[u] = sdom[u] = dsu[u] = u;

        for(int v : g[u]){
            if(!id[v])
                prep(v), down[u].push_back(v);
            gt[v].push_back(u);
        } // 4d7944
    } // 4351b9
    int fnd(int u, int flag = 0){
        if(u == dsu[u]) return u;
        int v = fnd(dsu[u], 1), b = label[ dsu[u] ];
        if(id[ sdom[b] ] < id[ sdom[ label[u] ] ])
            label[u] = b;
        dsu[u] = v;
        return flag ? v : label[u];
    } // d64927
}; // d7c79f
```

## 5.5 Cliques

**MaximalCliques.h**  
**Description:** Ruins a callback for all maximal cliques in a graph (given as a symmetric bitset matrix; self-edges not allowed). Callback is given a bitset representing the maximal clique.  
**Time:**  $\mathcal{O}\left(3^{n/3}\right)$ , much faster for sparse graphs

b0d5b1, 12L, 2m10s



```
typedef bitset<128> B;
template<class F>
void cliques(vector<B>& eds, F f, B P = ~B(), B X={}, B R={}) {
    if (!P.any()) { if (!X.any()) f(R); return; }
    auto q = (P | X)._Find_first();
    auto cands = P & ~eds[q];
    rep(i,0,sz(eds)) if (cands[i]) {
        R[i] = 1;
        cliques(eds, f, P & eds[i], X & eds[i], R);
        R[i] = P[i] = 0; X[i] = 1;
    } // 181f8f
} // c9dc5f
```

MaximumClique.h

**Description:** Quickly finds a maximum clique of a graph (given as symmetric bitset matrix; self-edges not allowed). Can be used to find a maximum independent set by finding a clique of the complement graph.  
**Time:** Runs in about 1s for n=155 and worst case random graphs (p=.90). Runs faster for sparse graphs.

```
typedef vector<bitset<200>> vb;
struct Maxclique {
    double limit=0.025, pk=0;
    struct Vertex { int i, d=0; };
    typedef vector<Vertex> vv;
    vb e;
    vv V;
    vector<vi> C;
    vi qmax, q, S, old;
    void init(vv& r) {
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto j : r) v.d += e[v.i][j.i];
        sort(all(r), [](auto a, auto b) { return a.d > b.d; });
        int mxD = r[0].d;
        rep(i,0,sz(r)) r[i].d = min(i, mxD) + 1;
    } // 7c428e
    void expand(vv& R, int lev = 1) {
        S[lev] += S[lev - 1] - old[lev];
        old[lev] = S[lev - 1];
        while (sz(R)) {
            if (sz(q) + R.back().d <= sz(qmax)) return;
            q.push_back(R.back().i);
            vv T;
            for(auto v:R) if (e[R.back().i][v.i]) T.push_back({v.i});
            if (sz(T)) {
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max(sz(qmax) - sz(q) + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) {
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk + 1].clear();
                    if (k < mnk) T[j++] .i = v.i;
                    C[k].push_back(v.i);
                } // 3221ac
                if (j > 0) T[j - 1].d = 0;
                rep(k,mnk,mxk + 1) for (int i : C[k])
                    T[j].i = i, T[j++].d = k;
                expand(T, lev + 1);
            } else if (sz(q) > sz(qmax)) qmax = q; // 2a0537
            q.pop_back(), R.pop_back();
        } // 87639b
    } // f0a49d
    vi maxClique() { init(V), expand(V); return qmax; }
    Maxclique(vb conn) : e(conn), C(sz(e)+1), S(sz(C)), old(S) {
        rep(i,0,sz(e)) V.push_back({i});
    } // 36accb
```

```
}; // b63641
```

5.6 Trees

CompressTree.h

**Description:** Given a rooted tree and a subset S of nodes, compute the minimal subtree that contains all the nodes by adding all (at most  $|S| - 1$ ) pairwise LCA's and compressing edges. Returns a list of (par, orig.index) representing a tree rooted at 0. The root points to itself.  
**Time:**  $\mathcal{O}(|S| \log |S|)$

```
struct LCA {
    vi time;
    int lca(int a, int b) { return -1; }
}; // 32d408

typedef vector<pair<int, int>> vpi;
vpi compressTree(LCA& lca, const vi& subset) {
    static vi rev; rev.resize(sz(lca.time));
    vi li = subset, &T = lca.time;
    auto cmp = [&](int a, int b) { return T[a] < T[b]; };
    sort(all(li), cmp);
    int m = sz(li)-1;
    rep(i,0,m) {
        int a = li[i], b = li[i+1];
        li.push_back(lca.lca(a, b));
    } // 677c62
    sort(all(li), cmp);
    li.erase(unique(all(li)), li.end());
    rep(i,0,sz(li)) rev[li[i]] = i;
    vpi ret = {pii(0, li[0])};
    rep(i,0,sz(li)-1) {
        int a = li[i], b = li[i+1];
        ret.emplace_back(rev[lca.lca(a, b)], b);
    } // 5efe90
    return ret;
} // 83c9a2

HLD.h
Description: Does the same thing as a LazySegment tree, but on a tree and with an extra log factor to the complexity. VAL_EDGES determines if the values are stored in edges. If so, store the value of each edge on its child vertex.
Time:  $\mathcal{O}((\log N)^2)$  for range operations,  $\mathcal{O}(\log(n))$  for point operations
"../data-structures/LazySegmentTree.h" b7947a, 53L, 9m40s

template<class S, bool VAL_EDGES>
struct HLD {
    using T = typename S::T; using L = typename S::L;
    int n,tim=0;
    vector<vi> g;
    vi p,siz,hd,ti;
    SegBeats<S> seg, segi;
    HLD(vector<vi> ag) : n(sz(ag)), g(ag), p(n), siz(n,1), hd(n), ti(n),
        seg(n), segi(n) { dfs(0); dfs2(0); }
    #warning fix n to be a power of two
    void dfs(int v) {
        for (int& u : g[v]) {
            g[u].erase(find(all(g[u]), v));
            p[u]=v;
            dfs(u);
            siz[v] += siz[u];
            if (siz[u] > siz[g[v][0]]) swap(u, g[v][0]);
        } // 7b54be
    } // a86bcf
    void dfs2(int v) {
        ti[v]=tim++;
        for (int u : g[v]) {
```

```
            hd[u] = (u == g[v][0] ? hd[v] : u);
            dfs2(u);
        } // 79707a
    } // 816069
    template <class B, class C> // ops are [l, r)
    void process(int u, int v, B opl, C op2) {
        for (;;) {
            if (hd[u] == hd[v]) break;
            if (ti[u] < ti[v]) op2(ti[hd[v]], ti[v]+1, v=p[hd[v]]);
            else opl(ti[hd[u]], ti[u]+1, u=p[hd[u]]);
        } // 1cb33f
        if (ti[u] < ti[v]) op2(ti[u]+VAL_EDGES, ti[v]+1);
        else opl(ti[v]+VAL_EDGES, ti[u]+1);
    } // 6a6a62
    T query(int u, int v) { // inclusive on both ends
        T vu = S::id, vv = S::id;
        process(u,v,
            [&](int l, int r){ vu = S::op(vu, segi.query(n-r,n-1)); },
            [&](int l, int r){ vv = S::op(seg.query(l,r), vv); } );
        return S::op(vu,vv);
    } // 04b81f
    void update(L val, int u, int v) {
        process(u,v,
            [&](int l, int r){ segi.update(val,n-r,n-1); },
            [&](int l, int r){ seg.update(val,l,r); });
    } // 8e0c4e
    void update(L val, int u) {
        seg.update(val,ti[u], ti[u]+1);
        segi.update(val,n-ti[u]-1, n-ti[u]);
    } // 6fea94
}; // 14746f
```

LinkCutTree.h

**Description:** Represents a forest of unrooted trees. You can add and remove edges (as long as the result is still a forest), and check whether two nodes are in the same tree.  
**Time:** All operations take amortized  $\mathcal{O}(\log N)$ .

```
struct Node { // Splay tree. Root's pp contains tree's parent.
    Node *p = 0, *pp = 0, *c[2];
    bool flip = 0;
    Node() { c[0] = c[1] = 0; fix(); }
    void fix() {
        if (c[0]) c[0]->p = this;
        if (c[1]) c[1]->p = this;
        // (+ update sum of subtree elements etc. if wanted)
    } // 454758
    void pushFlip() {
        if (!flip) return;
        flip = 0; swap(c[0], c[1]);
        if (c[0]) c[0]->flip ^= 1;
        if (c[1]) c[1]->flip ^= 1;
    } // 0cc949
    int up() { return p ? p->c[1] == this : -1; }
    void rot(int i, int b) {
        int h = i ^ b;
        Node *x = c[i], *y = b == 2 ? x : x->c[h], *z = b ? y : x;
        if ((y->p = p)) p->c[up()] = y;
        c[i] = z->c[i ^ 1];
        if (b < 2) {
            x->c[h] = y->c[h ^ 1];
            y->c[h ^ 1] = x;
        } // 1a82cf
        z->c[i ^ 1] = this;
        fix(); x->fix(); y->fix();
        if (p) p->fix();
        swap(pp, y->pp);
    } // 1cf643
```

```
void splay() {
    for (pushFlip(); p; ) {
        if (p->p) p->p->pushFlip();
        p->pushFlip(); pushFlip();
        int c1 = up(), c2 = p->up();
        if (c2 == -1) p->rot(c1, 2);
        else p->p->rot(c2, c1 != c2);
    } // e639f4
} // bfb1f7
Node* first() {
    pushFlip();
    return c[0] ? c[0]->first() : (splay(), this);
} // 67f9a1
}; // 225109
struct LinkCut {
    vector<Node> node;
    LinkCut(int N) : node(N) {}
    void link(int u, int v) { // add an edge (u, v)
        assert(!connected(u, v));
        makeRoot(&node[u]);
        node[u].pp = &node[v];
    } // 60799e
    void cut(int u, int v) { // remove an edge (u, v)
        Node *x = &node[u], *top = &node[v];
        makeRoot(top); x->splay();
        assert(top == (x->pp ? x->c[0]));
        if (x->pp) x->pp = 0;
        else {
            x->c[0] = top->p = 0;
            x->fix();
        } // 8acbe8
    } // a58ec7
    bool connected(int u, int v) { // are u, v in the same tree?
        Node* nu = access(&node[u])->first();
        return nu == access(&node[v])->first();
    } // b80a22
    void makeRoot(Node* u) {
        access(u);
        u->splay();
        if (u->c[0]) {
            u->c[0]->p = 0;
            u->c[0]->flip ^= 1;
            u->c[0]->pp = u;
            u->c[0] = 0;
            u->fix();
        } // 586a65
    } // 74c908
    Node* access(Node* u) {
        u->splay();
        while (Node* pp = u->pp) {
            pp->splay(); u->pp = 0;
            if (pp->c[1]) {
                pp->c[1]->p = 0; pp->c[1]->pp = pp; } // 1ccdfe
            pp->c[1] = u; pp->fix(); u = pp;
        } // b10f33
        return u;
    } // 4ac291
}; // ceab83
```

**DirectedMST.h**  
**Description:** Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.  
**Time:**  $\mathcal{O}(E \log V)$

|                                         |                     |
|-----------------------------------------|---------------------|
| ../data-structures/UnionFindRollback.h" | 39e620, 58L, 10m20s |
|-----------------------------------------|---------------------|

```
struct Edge { int a, b; ll w; };
struct Node {
    Edge key;
    Node *l, *r;
```

```
ll delta;
void prop() {
    key.w += delta;
    if (l) l->delta += delta;
    if (r) r->delta += delta;
    delta = 0;
} // 0d348f
Edge top() { prop(); return key; }
}; // ab4902
Node *merge(Node *a, Node *b) {
    if (!a || !b) return a ? b;
    a->prop(), b->prop();
    if (a->key.w > b->key.w) swap(a, b);
    swap(a->l, (a->r = merge(b, a->r)));
    return a;
} // c5109e
void pop(Node* &a) { a->prop(); a = merge(a->l, a->r); }
pair<ll, vi> dmst(int n, int r, vector<Edge> &g) {
    RollbackUF uf(n);
    vector<Node*> heap(n);
    for (Edge e : g) heap[e.b] = merge(heap[e.b], new Node{e});
    ll res = 0;
    vi seen(n, -1), path(n), par(n);
    seen[r] = r;
    vector<Edge> Q(n), in(n, {-1,-1}), comp;
    deque<tuple<int, int, vector<Edge>>> cycs;
    rep(s,0,n) {
        int u = s, qi = 0, w;
        while (seen[u] < 0) {
            if (!heap[u]) return {-1,{};};
            Edge e = heap[u]->top();
            heap[u]->delta -= e.w, pop(heap[u]);
            Q[qi] = e, path[qi++] = u, seen[u] = s;
            res += e.w, u = uf.find(e.a);
            if (seen[u] == s) {
                Node* cyc = 0;
                int end = qi, time = uf.time();
                do cyc = merge(cyc, heap[w = path[--qi]]);
                while (uf.join(u, w));
                u = uf.find(u), heap[u] = cyc, seen[u] = -1;
                cycs.push_front({u, time, {&Q[qi], &Q[end]}});
            } // 00a339
        } // c8f0da
        rep(i,0,qi) in[uf.find(Q[i].b)] = Q[i];
    } // fa3c2c
    for (auto& [u,t,comp] : cycs) { // restore sol (optional)
        uf.rollback(t);
        Edge inEdge = in[u];
        for (auto& e : comp) in[uf.find(e.b)] = e;
        in[uf.find(inEdge.b)] = inEdge;
    } // 4f9b56
    rep(i,0,n) par[i] = in[i].a;
    return {res, par};
} // efa3a4
```

**SmallToLarge.h**  
**Description:** good small to large  
**Time:**  $\mathcal{O}(N \log N)$

|  |                    |
|--|--------------------|
|  | 19475e, 26L, 5m30s |
|--|--------------------|

```
void small_to_large(vector<vector<int>> &g){
    int n = sz(g);
    int cans = 0;
    auto add = [&](int v){ cans += v; };
    auto del = [&](int v){ cans -= v; };
    vector ans(n, 0), big(n, -1), sub(n, 1), pos(n, 0), pre(0, 0)
    ;
    auto dfspre = [&](auto&& rec, int v, int p)->void {
        pos[v] = sz(pre); pre.push_back(v);
        for(int f : g[v])if (f != p){
```

```
            rec(rec, f, v);
            sub[v] += sub[f];
            if (big[v] == -1 or sub[f] > sub[big[v]])big[v] = f;
        } // 2f5782
    }; // 125df2
    dfspre(dfspre, 0, 0);
    auto dfs = [&](auto&& rec, int v, int p, int keep)->void {
        int b = big[v];
        for(int f : g[v])if (f != p and f != b)rec(rec, f, v, 0);
        if (b != -1)rec(rec, b, v, 1);
        add(v);
        for(int f : g[v])if (f != p and f != b)rep(i,pos[f],pos[f]+
            sub[f])add(pre[i]);
        ans[v] = cans;
        if (not keep)rep(i,pos[v],pos[v]+sub[v])del(pre[i]);
    }; // 022003
    dfs(dfs, 0, 0, 1);
} // 19475e
```

## Data structures (6)

### 6.1 Collections

**HashMap.h**  
**Description:** Hash map with mostly the same API as unordered\_map, but ~3x faster. Uses 1.5x memory. Initial capacity must be a power of 2 (if provided).

|  |                |
|--|----------------|
|  | d77092, 7L, 2m |
|--|----------------|

```
#include <bits/extc++.h>
// To use most bits rather than just the lowest ones:
struct chash { // large odd number for C
    const uint64_t C = 1l(4e18 * acos(0)) | 71;
    ll operator()(ll x) const { return __builtin_bswap64(x*C); }
}; // cdd37e
__gnu_pbds::gp_hash_table<ll,int,chash> h({}, {}, {}, {}, {1<<16});
```

**OrderedSet.h**  
**Description:** ordered/indexed set and multiset. Bad constant factor, works only in Linux. All erase/find/search functions assume the target exists.  
**Memory:**  $\mathcal{O}(N)$   
**Time:**  $\mathcal{O}(\log N)$  operations

|                                                 |                    |
|-------------------------------------------------|--------------------|
| <bits/extc++.h> //include it before any defines | b8c30a, 15L, 4m20s |
|-------------------------------------------------|--------------------|

```
using namespace __gnu_pbds;
template<class T, class B = null_type> using ordered_set = tree
<T, B, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
```

```
template<class T>
struct ordered_multiset{
    ordered_set<pair<T, int>> o; int c;
    ordered_multiset():c(0){}
    unsigned order_of_key(T x){return o.order_of_key({x, -1});}
    const T* find_by_order(int p){return &(*o.find_by_order(p)).
        first;}
    void insert(T x){o.insert({x, c++});}
    void erase(T x){o.erase(o.lower_bound({x, 0}));}
    unsigned size(){return o.size();}
    const T* lower_bound(T x){return &(*o.lower_bound({x, 0})).
        first;}
    const T* upper_bound(T x){return &(*o.upper_bound({x, c})).
        first;}
}; // d4e54f
```

UnionFindRollback.h

**Description:** Disjoint-set data structure with undo. If undo is not needed, skip st, time() and rollback().

**Usage:** int t = uf.time(); ...; uf.rollback(t);

**Time:**  $\mathcal{O}(\log(N))$

```
struct RollbackUF {
    vi e; vector<pii> st;
    RollbackUF(int n) : e(n, -1) {}
    int size(int x) { return -e[find(x)]; }
    int find(int x) { return e[x] < 0 ? x : find(e[x]); }
    int time() { return sz(st); }
    void rollback(int t) {
        for (int i = time(); i --> t;)
            e[st[i].first] = st[i].second;
        st.resize(t);
    } // 30bb61
    bool join(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (e[a] > e[b]) swap(a, b);
        st.push_back({a, e[a]});
        st.push_back({b, e[b]});
        e[a] += e[b]; e[b] = a;
        return true;
    } // 6c709f
}; // de4ad0
```

6.2 Line Queries

StaticCHT.h

**Description:** static CHT - add must be ordered by slope (a), queries by x. **Time:** amortized  $\mathcal{O}(1)$ .

```
struct CHT {
    int it; vector<ll> a, b;
    CHT():it(0){}
    ll eval(int i, ll x){return a[i]*x + b[i];}
    bool useless(){
        int sz = a.size();
        int r = sz-1, m = sz-2, l = sz-3;
#warning careful with overflow!
        return (b[l] - b[r])*a[m] - a[l]) <=
            (b[l] - b[m])*a[r] - a[l]);
    } // 848c9b
    void add(ll A, ll B){
        if(sz(a) && A==a.back()){
            if(B<=b.back())return;
            a.pop_back(); b.pop_back();
        } // 915a14
        a.push_back(A); b.push_back(B);
        while (!a.empty()){
            if ((a.size() < 3) || !useless()) break;
            a.erase(a.end() - 2);
            b.erase(b.end() - 2);
        } // b21fc8
        it = min(it, int(a.size()) - 1);
    } // 9ef59c
    ll query(ll x){
        if (sz(a) == 0)return -oo;
        while (it+1 < a.size()){
            if (eval(it+1, x) >= eval(it, x)) it++;
            else break;
        } // 27ce3f
        return eval(it, x);
    } // adab31
}; // 8331db
```

LineContainer.h

**Description:** Dynamic CHT where you can add lines of the form ax+b, and query maximum values at points x. For doubles, use inf = 1/.0, floordiv(a,b) = a/b

**Time:**  $\mathcal{O}(\log N)$

```
ll floordiv(ll a, ll b) {
    return a / b - ((a ^ b) < 0 && a % b);
} // 7e2711
struct Line {
    mutable ll a, b, p;
    bool operator<(const Line& o) const { return a < o.a; }
    bool operator<(ll x) const { return p < x; }
}; // 469928
struct LineContainer : multiset<Line, less<>> {
    bool isect(iterator i, iterator j) {
        if (j == end()) return i->p = oo, 0;
        if (i->a == j->a) i->p = i->b > j->b ? oo : -oo;
        else i->p = floordiv(j->b - i->b, i->a - j->a);
        return i->p >= j->p;
    } // 978e9c
    void add(ll a, ll b) {
        auto k = insert({a, b, 0}), j = k++, i = j;
        while (isect(j, k)) k = erase(k);
        if (i != begin() && isect(--i, j)) isect(i, j = erase(j));
        while ((j = i) != begin() && (--i)->p >= j->p)
            isect(i, erase(j));
    } // 99e4d6
    ll query(ll x) {
        if (empty())return -oo;
        auto l = *lower_bound(x);
        return l.a * x + l.b;
    } // 22865e
}; // 35b669
```

LiChao.h

**Description:** Compressed Li Chao tree for max queries. Give all query x-coordinates in xs. Lines are added on [l,r].

**Time:**  $\mathcal{O}(\log^2 N)$  add,  $\mathcal{O}(\log N)$  query.

```
template<class T> struct LiChao {
    static constexpr T NINF = numeric_limits<T>::lowest();
    struct Line {
        T a=0,b=NINF;
        T operator()(T x) { return a*x+b; }
    }; // 62e8ca
    int n; vector<T> xs; vector<Line> t;
    LiChao(vector<T> _xs):xs(_xs) {
        sort(all(xs));
        xs.erase(unique(all(xs)),xs.end());
        n = sz(xs);
        t.resize(4*n);
    } // f827f2
    void add(Line v, int p, int l, int r) {
        int m=(l+r)/2;
        bool L=v(xs[l])>t[p](xs[l]), M=v(xs[m])>t[p](xs[m]);
        if (M) swap(v,t[p]);
        if (r-l==1) return;
        if (L!=M) add(v,2*p+1,m,r);
        else add(v,2*p+1,m,r);
    } // cc3a96
    void add(Line v, int ql, int qr, int p, int l, int r) {
        if (qr<=l or r<=ql) return;
        if (ql<=l and r<=qr) return add(v,p,l,r);
        int m=(l+r)/2;
        add(v,ql,qr,2*p+1,m); add(v,ql,qr,2*p+1,m,r);
    } // 6f709f
    void add(T a, T b, T l, T r) {
        int li=lower_bound(all(xs),l)-xs.begin();
```

```
int ri=upper_bound(all(xs),r)-xs.begin();
if (li<ri) add({a,b,li,ri,l,0,n});
} // 34dfcd
T query(T x, int i, int p, int l, int r) {
    T ans=t[p](x);
    if (r-l==1) return ans;
    int m=(l+r)/2;
    T best = i < m ? query(x,i,2*p,l,m) : query(x,i,2*p+1,m,r);
    return max(ans, best);
} // fe58ac
T query(T x) {
    int i=lower_bound(all(xs),x)-xs.begin();
    return query(x,i,l,0,n);
} // 438b30
}; // 14b532
```

6.3 Range Queries

Spec.h

**Description:** Spec for Seg Beats. It's similar to other data structures. If id is not contexpr, declare "BeatsSpec::T BeatsSpec:: id;" globally and initialize "BeatsSpec::id" inside main

**Time:** varies

```
struct node{int max1,max2,maxc; ll sum;};
struct BeatsSpec{ //chmin sum
    using T = node; using L = int;
    static constexpr T id = node{-oo,-oo,0,0};
    static T op(T a, T b){
        node n;
        if (a.max1 > b.max1){
            n.max1 = a.max1;
            n.max2 = max(a.max2, b.max1);
            n.maxc = a.maxc;
        } // 60044a
        else if (a.max1 == b.max1){
            n.max1 = a.max1;
            n.max2 = max(a.max2, b.max2);
            n.maxc = a.maxc+b.maxc;
        } // 89a062
        else{
            n.max1 = b.max1;
            n.max2 = max(b.max2, a.max1);
            n.maxc = b.maxc;
        } // dbf86a
        n.sum=a.sum+b.sum;
        return n;
    } // 7211b8
    static T ch(T a, L b, int l, int r){
        if (a.max1 > b)a.sum -= (ll)(a.max1-b)*a.maxc, a.max1 = b;
        return a;
    } // 54e66b
    static L cmp(L a, L b){return min(a,b);}
    static bool brk(L a, T b){return b.max1 <= a;}
    static bool tag(L a, T b){return b.max2 < a;}
}; // 9e405c
```

BIT.h

**Description:** BIT/Fenwick Tree, requires abelian group (T, id, op, inv). For  $N = 10^7$ , allocate h(n)+1 and swap all access bit[i] for bit[h(i)]

**Time:**  $\mathcal{O}(\log N)$ .

```
#define chf(a, f, b) a = f(a, b)
//int h(x){return x+(x>>10);} for N = 10^7$
template<class S> struct BIT{
    using T = typename S::T;
    int n; vector<T> bit;
    int lb(int x){return x&(-x);}
    BIT(vector<T> v):n(sz(v)),bit(1+n,S::id){
```

```
rep(i,1,n+1){
    chf(bit[i], S::op, v[i-1]);
    if (i+lb(i)<=n)chf(bit[i+lb(i)], S::op, bit[i]);
} // 41e113
} // 484915
void add(T x, int p){for(p++;p<n;p+=lb(p)) chf(bit[p],S::op,x);}
T query(int l, int r){
    T lv=S::id, rv=S::id; r++;
    for(;r>=l;r-=lb(r)) chf(rv,S::op,bit[r]);
    for(;l>=1;l-=lb(l)) chf(lv,S::op,bit[l]);
    return S::op(rv,S::inv(lv));
} // 990a13
}; // 4b6a8d
```

**Dist.h**  
**Description:** Disjoint Sparse Table Requires Monoid (op, id)  
**Memory:**  $\mathcal{O}(N \log N)$   
**Time:**  $\mathcal{O}(1)$

```
#define repinv(i, a, b) for(int i = (a); i >= (b); i--)
template<class S> struct DiST{
    using T = typename S::T;
    int n, h; vector<vector<T>> t;
    int lg(unsigned x){return bit_width(x)-1;}
    DiST(vector<T> v): n(bit_ceil(v.size())),
        h(lg(n)), t(h, vector<T>(n)){
        v.resize(n, S::id);
        if (n == 1)t.pb(v);
        for(int d = 0, s = 1; d < h; d++, s *= 2)
            for(int m = s; m < n; m += 2*s){
                t[d][m] = v[m]; t[d][m-1] = v[m-1];
                rep(i, m+1, m+s)t[d][i] = S::op(t[d][i-1], v[i]);
                repinv(i, m-2, m-s)t[d][i] = S::op(v[i], t[d][i+1]);
            } // 3b44fe
        } // 2329d7
    T query(int l, int r){
        if (l==r)return t[0][l];
        int k = lg(l^r);
        return S::op(t[k][l], t[k][r]);
    } // 07c10a
}; // 8eff81
```

**SparseTable.h**  
**Description:** Sparse Table Requires Idempotent Monoid S (op, id)  
**Memory:**  $\mathcal{O}(n \log n)$   
**Time:**  $\mathcal{O}(1)$  query,  $\mathcal{O}(n \log n)$  build

```
template<class S> struct SpTable{
    using T = typename S::T;
    int n; vector<vector<T>> tab;
    int lg(signed x){return __builtin_clz(1)-__builtin_clz(x);}
    SpTable(vector<T> v):n(sz(v)),tab(1+lg(n),vector<T>(n,S::id)){
        rep(i,0,n)tab[0][i] = v[i];
        rep(i,0,lg(n))rep(j,0,n-(1<<i))
            tab[i+1][j] = S::op(tab[i][j], tab[i][j+(1<<i)]);
    } // c105d7
    T query(int l, int r){
        int k = lg(++r-1);
        return S::op(tab[k][l], tab[k][r-(1<<k)]);
    } // e06689
}; // 276609
```

**Treap.h**  
**Description:** Implicit Treap. Dynamic array with insert/erase, lazy range updates, queries and reversals. Ranges are inclusive. Operations are described at the end. S requires T,L,id,op,ch,cmp,rev (what happens to range val when reversed). ch always receives [0,len-1]. Lazy operations must commute with reversions.  
**Time:**  $\mathcal{O}(\log N)$  expected per operation.

```
2e1a94, 84L, 14m10s
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

template<class S> struct Treap {
    using T=typename S::T; using L=typename S::L;
    struct N {
        int l=0, r=0,c=0; unsigned y=0;
        T v=S::id, ac=S::id; L lz{};
        bool ig=1,rv=0;
        N(int C=0, T V=S::id, unsigned Y=0):c(C),y(Y),v(V),ac(V){}
    }; // a02fb4
    vector<N> t; int rt=0;
    Treap():t(1){}
    int node(T v){return t.emplace_back(l,v,rng()), sz(t)-1;}
    void apply(int p,L v){
        if(!p)return;
        t[p].v=S::ch(t[p].v,v,0,0);
        t[p].ac=S::ch(t[p].ac,v,0,t[p].c-1);
        t[p].lz=t[p].ig?v:S::cmp(t[p].lz,v); t[p].ig=0;
    } // e37416
    void flip(int p){
        if(!p)return;
        swap(t[p].l,t[p].r);
        t[p].ac=S::rev(t[p].ac);
        t[p].rv^=1;
    } // b92f4f
    void push(int p){
        if(t[p].rv) flip(t[p].l),flip(t[p].r),t[p].rv=0;
        if(!t[p].ig)apply(t[p].l,t[p].lz),apply(t[p].r,t[p].lz),t[p].ig=1;
    } // 42a806
    void pull(int p){
        t[p].c=t[t[p].l].c+t[t[p].r].c+1;
        t[p].ac=S::op(S::op(t[t[p].l].ac,t[p].v),t[t[p].r].ac);
    } // 1bc654
    pair<int,int> split(int p,int k){ // first k go left
        if(!p)return {};
        push(p);
        if(t[t[p].l].c>=k){
            auto [a,b]=split(t[p].l,k);
            t[p].l=b; pull(p); return {a,p};
        } // f522e3
        auto [a,b]=split(t[p].r,k-t[t[p].l].c-1);
        t[p].r=a; pull(p); return {p,b};
    } // a5d263
    int merge(int a,int b){
        if(!a||!b)return a?a:b;
        if(t[a].y>t[b].y){
            push(a); t[a].r=merge(t[a].r,b); pull(a); return a;
        } // 77ec5b
        push(b); t[b].l=merge(a,t[b].l); pull(b); return b;
    } // c5d1b2
}; // 6195b6
```

```
/* Additional optional operations:
Treap(vector<T> a={}):t(1){
    t.reserve(sz(a)+1);
    for(T x:a)rt=merge(rt,node(x));
} // 6d6612
array<int,3> cut(int l,int r){
    auto [a,c]=split(rt,r+1);
    auto [x,b]=split(a,l);
```

```
return {x,b,c};
} // dbc1d0
void join(array<int,3> a){rt=merge(merge(a[0],a[1]),a[2]);}
void insert(int p,T v){
    auto [a,b]=split(rt,p);
    rt=merge(merge(a,node(v)),b);
} // 46fa14
void erase(int l,int r){
    auto a=cut(l,r);
    rt=merge(a[0],a[2]);
} // bde6f5
void update(L v,int l,int r){
    auto a=cut(l,r); apply(a[1],v); join(a);
} // 0f7e99
void reverse(int l,int r){
    auto a=cut(l,r); flip(a[1]); join(a);
} // 2425ca
T query(int l,int r){
    auto a=cut(l,r);
    T ans=t[a[1]].ac; join(a); return ans;
} // a1eeba
T get(int p){return query(p,p);}
int size(){return t[rt].c;}
*/
```

**Mo.h**  
**Description:** Answer interval or tree path queries. Includes interval version without deletion. If values are on tree edges, change step to add/remove the edge (a,c) and remove the initial add call (but keep in).  
**Time:**  $\mathcal{O}(N\sqrt{Q})$

```
1f1cfd, 76L, 15m
void add(int i, int end) { i+=end; }
void del(int i, int end) { i-=end; }
int calc() { return 0; }

vi mo(vector<pii> Q) {
    int L = 0, R = -1, blk = 450; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s;
    #define K(x) pii(x.first/blk, x.second ^ -(x.first/blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int i, int j){ return K(Q[i]) < K(Q[j]); });
    for (int qi : s) {
        auto [l, r] = Q[qi];
        while (L > l) add(--L, 0);
        while (R < r) add(++R, 1);
        while (L < l) del(L++, 0);
        while (R > r) del(R--, 1);
        res[qi] = calc();
    } // 241d86
    return res;
} // d18826
```

```
vi moTree(vector<array<int, 2>> Q, vector<vi>& ed, int root=0){
    int N = sz(ed), pos[2] = {}, blk = 350; // ~N/sqrt(Q)
    vi s(sz(Q)), res = s, I(N), L(N), R(N), in(N), par(N);
    add(root, 0), in[root] = 1;
    auto dfs = [&](int x, int p, int dep, auto& f) -> void {
        par[x] = p;
        L[x] = N;
        if (dep) I[x] = N++;
        for (int y : ed[x]) if (y != p) f(y, x, !dep, f);
        if (!dep) I[x] = N++;
        R[x] = N;
    }; // 329c88
    dfs(root, -1, 0, dfs);
    #define K(x) pii(I[x[0]] / blk, I[x[1]] ^ -(I[x[0]] / blk & 1))
    iota(all(s), 0);
    sort(all(s), [&](int a, int b){ return K(Q[a]) < K(Q[b]); });
    for (int qi : s) rep(end,0,2) {
```

```
int &a = pos[end], b = Q[qi][end], i = 0;
#define step(c) { if (in[c]) { del(a, end); in[a] = 0; } \
                else { add(c, end); in[c] = 1; } a = c; } //
                3839ba
while (!(L[b] <= L[a] && R[a] <= R[b]))
    I[i++] = b, b = par[b];
while (a != b) step(par[a]);
while (i--) step(I[i]);
if (end) res[qi] = calc();
} // c880be
return res;
} // f45ee8
```

```
vector<int> mo_no_deletion(vector<pii>& qs, int n){
    int q = sz(qs), sq = (int)sqrt(q)+1, blk = (n+sq+1)/sq;
    vector<vi> o((n+blk-1)/blk);
    rep(i,0,q)o[qs[i].first/blk].pb(i);
    for(auto& vq : o)sort(all(vq), [&](int i, int j){return qs[i]
        ].second < qs[j].second;});
    vector<int> ans(q);
    rep(i,0,sz(o)){
        auto& vq = o[i];
        int l = blk*i + blk, r = l-1;
        // prepare to answer queries
        for(int qi : vq){
            auto [ql, qr] = qs[qi];
            if (qr <= l){ //if it does not extrapolate
                rep(j,ql,qr+1)add(j, 1); //solving manually
                continue;
            } // b94913
            while(r < qr)add(++r, 1);
            int ml = l; //we will move l manually
            // prepare checkpoint
            while(ml > ql)add(--ml, 0);
            ans[qi] = calc();
            // revert checkpoint: discard changes made by moving l
        } // 6569ca
    } // 09ff2d
    return ans;
} // cf095f
```

6.3.1 Segment Tree

SegmentTree.h

Description: Iterative SegTree

Time:  $\mathcal{O}(\log N)$

```
template<class S> struct SegTree{
    using T = typename S::T;
    int n; vector<T> seg;
    SegTree(int s):n(s),seg(2*n,S::id){}
    void update(T x, int i){
        for(seg[i+=n]=x;i/=2;)
            seg[i] = S::op(seg[2*i], seg[2*i+1]);
    } // 781838
    T query(int l, int r){
        T vl=S::id, vr=S::id;
        for(l+=n,r+=n+1;l<r;l/=2,r/=2){
            if (l&l)vl = S::op(vl, seg[l++]);
            if (r&l)vr = S::op(seg[--r], vr);
        } // ea3c70
        return S::op(vl, vr);
    } // 524637
}; // 6263bc
```

LazySegmentTree.h

Description: Lazy Seg. Can be transformed into Seg Beats by uncommenting conditions.

Time:  $\mathcal{O}(\log N * (ch + cmp))$ .

```
template<class S> struct SegBeats{
    using T = typename S::T; using L = typename S::L;
    int n; vector<T> seg; vector<L> lz; vector<bool> ig;
    SegBeats(unsigned s):n(bit_ceil(s)),seg(2*n,S::id),lz(2*n),ig
        (2*n,1){}
    void apply(int p, L v, int l, int r){
        seg[p] = S::ch(seg[p],v,l,r-1);
        if (r-1>l)lz[p] = ig[p] ? v : S::cmp(lz[p], v), ig[p] = 0;
    } // ada588
    int prop(int p, int l, int r){
        int m = (l+r)/2;
        if (not ig[p])ig[p] = 1,
            apply(2*p, lz[p], l, m), apply(2*p+1, lz[p], m, r);
        return m;
    } // 3caddda
    void update(L v, int l, int r){return update(v,l,r+1,1,0,n);}
    void update(L v, int lq, int rq, int no, int lx, int rx){
        if (rq <= lx or rx <= lq /*or S::brk(v,seg[no])*/)return;
        if (lq <= lx and rx <= rq /*and S::tag(v,seg[no])*/)return
            apply(no, v, lx, rx);
        int mx = prop(no,lx,rx);
        update(v,lq,rq,2*no,lx,mx); update(v,lq,rq,2*no+1,mx,rx);
        seg[no] = S::op(seg[2*no],seg[2*no+1]);
    } // f11ac8
    T query(int l, int r){return query(l,r+1,1,0,n);}
    T query(int lq, int rq, int no, int lx, int rx){
        if (rq <= lx or rx <= lq)return S::id;
        if (lq <= lx and rx <= rq)return seg[no];
        int mx = prop(no,lx,rx);
        return S::op(query(lq,rq,2*no,lx,mx),query(lq,rq,2*no+1,mx,
            rx));
    } // 0ed921
}; // 1ad942
```

SparseSeg.h

Description: Sparse Seg (I is index type)

Time:  $\mathcal{O}(\log N)$

```
template<class I, class S> struct SparseSeg{
    using T = typename S::T;
    I n; vector<int> lc, rc; vector<T> val;
    int new_node(){
        int res = sz(lc);
        lc.pb(0); rc.pb(0); val.pb(S::id);
        return res;
    } // b4e987
    SparseSeg(I s): n(s){ new_node(); new_node(); }
    void update(T x, I p){return update(x, p, 1, 0, n);}
    void update(T x, I p, int i, I lx, I rx){
        if (rx-lx==1){val[i] = x; return;}
        I mx = (lx+(rx-lx)/2);
        int li = lc[i], ri = rc[i];
        if (p < mx){
            if (not li)li = lc[i] = new_node();
            update(x, p, li, lx, mx);
        } // 452d3b
        else{
            if (not ri)ri = rc[i] = new_node();
            update(x, p, ri, mx, rx);
        } // 9a7652
        val[i] = S::op(val[li], val[ri]);
    } // 1ed0c3
    T query(I l, I r){return query(l, r+1, 1, 0, n);}
    T query(I l, I r, int i, I lx, I rx){
```

```
if (r <= lx or rx <= l)return S::id;
if (l <= lx and rx <= r)return val[i];
I mx = (lx+(rx-lx)/2);
return S::op(query(l, r, lc[i], lx, mx), query(l, r, rc[i],
    mx, rx));
} // 61def7
}; // 31cd4a
```

LazySparseSeg.h

Description: Lazy Sparse Seg, similar to Lazy Seg (I is index type). Queries also create nodes

Time:  $\mathcal{O}(\log N * (ch + cmp))$ .

```
template<class I, class S> struct LazySparseSeg{
    using T = typename S::T; using L = typename S::L;
    I n; vector<int> lc, rc; vector<bool> ig;
    vector<T> val; vector<L> lz;
    int new_node(){
        int res = sz(lc);
        lc.pb(0); rc.pb(0); ig.pb(1);
        val.pb(S::id); lz.pb(L());
        return res;
    } // 81c523
    LazySparseSeg(I s): n(s){ new_node(); new_node(); }
    void apply(int i, L v, I lx, I rx){
        val[i] = S::ch(val[i], v, lx, rx-1);
        if (rx-lx>1)lz[i] = ig[i] ? v : S::cmp(lz[i], v), ig[i] =
            0;
    } // 0e5514
    I prop(int i, I lx, I rx){
        I mx = (lx+(rx-lx)/2);
        if (!lc[i])lc[i] = new_node(), rc[i] = new_node();
        if (not ig[i])ig[i] = 1,
            apply(lc[i], lz[i], lx, mx), apply(rc[i], lz[i], mx, rx);
        return mx;
    } // b3b231
    void update(L v, I l, I r){return update(v, l, r+1, 1, 0, n);
    }
    void update(L v, I l, I r, int i, I lx, I rx){
        if (r <= lx or rx <= l)return;
        if (l <= lx and rx <= r)return apply(i, v, lx, rx);
        I mx = prop(i, lx, rx);
        int li = lc[i], ri = rc[i];
        update(v, l, r, li, lx, mx); update(v, l, r, ri, mx, rx);
        val[i] = S::op(val[li], val[ri]);
    } // bf0ca8
    T query(I l, I r){return query(l, r+1, 1, 0, n);}
    T query(I l, I r, int i, I lx, I rx){
        if (r <= lx or rx <= l)return S::id;
        if (l <= lx and rx <= r)return val[i];
        I mx = prop(i, lx, rx);
        return S::op(query(l, r, lc[i], lx, mx), query(l, r, rc[i],
            mx, rx));
    } // 1ef417
}; // a37732
```

PersistentSeg.h

Description: Persistent Segment Tree. Call "init" to get the root!

Time:  $\mathcal{O}(\log N)$

```
template<class I, class S> struct PersistentSeg{
    using T = typename S::T;
    I n; vector<int> lc, rc; vector<T> val;
    int new_node(int l, int r, T x){
        int res = sz(lc);
        lc.pb(1); rc.pb(r); val.pb(x);
        return res;
    } // eb1f5d
```

```
int new_node(int l, int r){return new_node(l, r, S::op(val[l], val[r]));}
PersistentSeg(){ new_node(0,0,S::id); }
int init(I s){return n = s, new_node(0,0,S::id);}
int update(T x, I p, int root){return update(x, p, root, 0, n);}
int update(T x, I p, int i, I lx, I rx){
    if (rx-lx==1)return new_node(0,0,x);
    I mx = lx + (rx - lx) / 2;
    if (p < mx)return new_node(update(x, p, lc[i], lx, mx), rc[i]);
    else return new_node(lc[i], update(x, p, rc[i], mx, rx));
} // 7644dd
T query(I l, I r, int root){return query(l, r+1, root, 0, n);}
T query(I l, I r, int i, I lx, I rx){
    if (r <= lx or rx <= l)return S::id;
    if (l <= lx and rx <= r)return val[i];
    I mx = (lx+(rx-lx)/2);
    return S::op(query(l, r, lc[i], lx, mx), query(l, r, rc[i], mx, rx));
} // 61def7
}; // 807b80
```

**LazyPersistentSeg.h**  
**Description:** Persistent Lazy Sparse Segment Tree. Call "init" to get the root! Queries also create nodes.  
**Time:**  $\mathcal{O}(\log N * (ch + cmp))$

```
template<class I, class S> struct LazyPersistentSeg{
    using T = typename S::T;
    using L = typename S::L;
    I n; vector<int> lc, rc; vector<bool> ig;
    vector<T> val; vector<L> lz;
    int new_node(int l, int r){
        int res = sz(lc);
        lc.pb(l); rc.pb(r); ig.pb(1);
        val.pb(S::op(val[l], val[r])); lz.pb(L());
        return res;
    } // e66e23
    LazyPersistentSeg(){
        lc.pb(0); rc.pb(0); ig.pb(1);
        val.pb(S::id); lz.pb(L());
    } // ccb55c
    int init(I s){return n = s, new_node(0, 0);}
    int lazy_clone(int i, L v, I lx, I rx){
        int ni = new_node(lc[i], rc[i]);
        lz[ni] = ig[i] ? v : S::cmp(lz[i], v);
        ig[ni] = 0; val[ni] = S::ch(val[i], v, lx, rx-1);
        return ni;
    } // 56e9d8
    I prop(int i, I lx, I rx){
        I mx = lx + (rx - lx) / 2;
        if (not ig[i]){
            ig[i] = 1; if (rx - lx > 1)
                lc[i] = lazy_clone(lc[i], lz[i], lx, mx),
                rc[i] = lazy_clone(rc[i], lz[i], mx, rx);
            } // a69d31
        return mx;
    } // 827037
    int update(L v, I l, I r, int root){return update(v, l, r+1, root, 0, n);}
    int update(L v, I l, I r, int i, I lx, I rx){
        if (r <= lx or rx <= l)return i;
        if (l <= lx and rx <= r)return lazy_clone(i, v, lx, rx);
        I mx = prop(i, lx, rx);
        return new_node(update(v, l, r, lc[i], lx, mx), update(v, l+1, r, rc[i], mx, rx));
    } // 49ce75
```

```
T query(I l, I r, int root){return query(l, r+1, root, 0, n);}
T query(I l, I r, int i, I lx, I rx){
    if (r <= lx or rx <= l)return S::id;
    if (l <= lx and rx <= r)return val[i];
    I mx = prop(i, lx, rx);
    return S::op(query(l, r, lc[i], lx, mx), query(l, r, rc[i], mx, rx));
} // 1ef417
}; // 47805d
```

6.3.2 MDRQDS

**MPsum.h**  
**Description:** Multidimensional Psum Requires Abelian Group (op, inv, id)  
**Memory:**  $\mathcal{O}(N^D)$   
**Time:**  $\mathcal{O}(2^D)$

```
#define MAS template<class... As>
template<int D, class S> struct MPsum{
    using T = typename S::T;
    int n; vector<MPsum<D-1, S>> v;
    MAS MPsum(int s, As... ds):n(s+1),v(n,MPsum<D-1, S>(ds...)){}
    MAS void set(T x, int p, As... ps){v[p+1].set(x, ps...);}
    void push(MPsum& p){rep(i, 1, n)v[i].push(p.v[i]);}
    void init(){rep(i, 1, n)v[i].init(),v[i].push(v[i-1]);}
    MAS T query(int l, int r, As... ps){
        return S::op(v[r+1].query(ps...),S::inv(v[l].query(ps...)))
        ;
    } // eac6a8
}; // f3cfd
template<class S> struct MPsum<0, S>{
    using T = typename S::T;
    T val=S::id;
    void set(T x){val=x;}
    void push(MPsum& a){val=S::op(a.val,val);}
    void init(){}
    T query(){return val;}
}; // ee6e4e
```

**MSpTable.h**  
**Description:** Multidimensional Sparse Table Requires commutative idempotent monoid (T, id, op)  
**Memory:**  $\mathcal{O}((N \log N)^D)$   
**Time:**  $\mathcal{O}(2^D)$

```
#define MAS template<class...As>
template<int D, class S> struct MSpTable{
    using T = typename S::T;
    using isp = MSpTable<D-1, S>;
    int lg(signed x){return __builtin_clz(1)-__builtin_clz(x);}
    int n; vector<vector<isp>> tab;
    MAS MSpTable(int s, As... ds):n(s),
        tab(1+lg(n),vector<isp>(n,isp(ds...))){}
    MAS void set(T x, int p, As... ps){tab[0][p].set(x, ps...);}
    void join(MSpTable& a, MSpTable& b){
        rep(i, 0, 1+lg(n))rep(j, 0, n)
            tab[i][j].join(a.tab[i][j], b.tab[i][j]);
    } // 96d587
    void init(){
        rep(i, 0, n)tab[0][i].init();
        rep(i, 0, lg(n))rep(j, 0, n-(1<<i))
            tab[i+1][j].join(tab[i][j], tab[i][j+(1<<i)]);
    } // dc476b
    MAS T query(int l, int r, As... ps){
        int k = lg(r-1+1); r+=1-(1<<k);
```

```
return S::op(tab[k][l].query(ps...),tab[k][r].query(ps...))
        ;
    } // b8ce4c
}; // 0764a3

template<class S> struct MSpTable<0, S>{
    using T = typename S::T;
    T val=S::id;
    void set(T x){val=x;}
    void join(MSpTable& a, MSpTable& b){val=S::op(a.val,b.val);}
    void init(){}
    T query(){return val;}
}; // e7585b
```

**MSegTree.h**  
**Description:** Multidimensional SegTree Requires commutative Monoid (op, id)  
**Memory:**  $\mathcal{O}(N^D)$   
**Time:**  $\mathcal{O}((\log N)^D)$

```
#define MAS template<class... As>
template<int D, class S> struct MSegTree{
    using T = typename S::T;
    int n; vector<MSegTree<D-1, S>> seg;
    MAS MSegTree(int s, As... ds):n(s),seg(2*n,MSegTree<D-1, S>(ds...)){}
    MAS T get(int p, As... ps){return seg[p+n].get(ps...);}
    MAS void update(T x, int p, As... ps){
        seg[p+n].update(x, ps...);
        while(p/=2)seg[p].update(S::op(seg[2*p].get(ps...),seg[2*p+1].get(ps...)), ps...);
    } // 2c8b52
    MAS T query(int l, int r, As... ps){
        T lv=S::id,rv=S::id;
        for(l+=n,r+=n+1;l<r;l/=2,r/=2){
            if (l&1)lv = S::op(lv,seg[l++].query(ps...));
            if (r&1)rv = S::op(seg[--r].query(ps...),rv);
        } // 1569b6
        return S::op(lv,rv);
    } // bc7474
}; // da06ff
template<class S> struct MSegTree<0, S>{
    using T = typename S::T;
    T val=S::id;
    T get(){return val;}
    void update(T x){val=x;}
    T query(){return val;}
}; // 50402e
```

**CMBIT.h**  
**Description:** Compressed multidimensional BIT. All update points must be given beforehand. Queries are inclusive. K here is the number of update points.  
**Memory:**  $\mathcal{O}(K \log^D - 1)K$ .  
**Time:**  $\mathcal{O}(\log^D K)$  update,  $\mathcal{O}((2 \log K)^D)$  query.

```
#define MAS template<class... As>
template<class I, int D, class S> struct CMBIT {
    using T=typename S::T; using P=array<I,D-1>;
    int n; vector<I> x; vector<CMBIT<I,D-1,S>> ft;
    int ilb(I i){return (int)(lower_bound(all(x),i)-x.begin());}
    int irb(I i){return (int)(upper_bound(all(x),i)-x.begin());}
    CMBIT(vector<array<I, D>> p){
        sort(all(p)); p.erase(unique(all(p)),p.end());
        x.reserve(sz(p));
        for(auto &a:p) if(x.empty() || x.back()!=a[0])x.pb(a[0]);
```

```
vector<vector<P>> ips(n = sz(x) + 1);
for(auto &a:p){
    P b; rep(j,1,D) b[j-1]=a[j];
    for(int i=ilb(a[0])+1;i<n;i+=i&-i)ips[i].pb(b);
} // 5e2514
ft.reserve(n); for(auto &y:ips) ft.eb(move(y));
} // 43e0d7
MAS void update(T v, I p, A... ps){
    for(int i=ilb(p)+1;i<n;i+=i&-i)ft[i].update(v,ps...);
} // b9d0b2
MAS T query(I l, I r, A... ps){
    T a=S::id,b=S::id;
    for(int R = irb(r);R;R=R&-R)b=S::op(b,ft[R].query(ps...));
    for(int L = ilb(l);L;L=L&-L)a=S::op(a,ft[L].query(ps...));
    return S::op(b,S::inv(a));
} // 963674
}; // ac3671
```

```
template<class I, class S> struct CMBIT<I,0,S> {
    using T=typename S::T;
    T v=S::id;
    CMBIT(vector<array<I,0>> ig) {}
    void update(T x){v=S::op(v,x);}
    T query() {return v;}
}; // ac559d
```

**CMSEG.h**  
Description: Compressed multidimensional SegTree. All update points must be given beforehand. Set updates and inclusive queries.  
Memory:  $\mathcal{O}\left(K\log^D(1)K\right)$ .  
Time:  $\mathcal{O}\left(\log^D K\right)$  update/query.

```
#define MAS template<class... A>
template<class I, int D, class S> struct CMSEG {
    using T=typename S::T; using P=array<I,D-1>;
    int n; vector<I> x; vector<CMSEG<I,D-1,S>> seg;
    int ilb(I i){return lower_bound(all(x),i)-x.begin();}
    int irb(I i){return upper_bound(all(x),i)-x.begin();}
    CMSEG(vector<array<I,D>> p){
        sort(all(p)); p.erase(unique(all(p)),p.end());
        x.reserve(sz(p));
        for(auto &a:p) if(x.empty() || x.back()!=a[0])x.pb(a[0]);
        vector<vector<P>> ips(2*(n=sz(x)));
        for(auto &a:p){
            P b; rep(j,1,D)b[j-1]=a[j];
            for(int i=ilb(a[0])+n;i;/=2)ips[i].pb(b);
        } // b4a19f
        seg.reserve(2*n); for(auto &y:ips)seg.eb(move(y));
    } // daffca
    MAS T get(I p,A... ps){
        int i=ilb(p);
        return i<n && x[i]==p ? seg[i+n].get(ps...) : S::id;
    } // 6aa8eb
    MAS void update(T v,I p,A... ps){
        int i=ilb(p)+n; seg[i].update(v,ps...);
        while(i/=2)seg[i].update(S::op(seg[2*i].get(ps...), seg[2*i+1].get(ps...)),ps...);
    } // d8166c
    MAS T query(I l,I r,A... ps){
        T lv=S::id,rv=S::id;
        for(int L=ilb(l)+n,R=irb(r)+n;L<R;L/=2,R/=2){
            if(L&1)lv=S::op(lv,seg[L+1].query(ps...));
            if(R&1)rv=S::op(seg[--R].query(ps...),rv);
        } // baadc3
        return S::op(lv,rv);
    } // bfe628
}; // 27a5de
```

```
template<class I, class S> struct CMSEG<I,0,S> {
    using T=typename S::T;
    T v=S::id;
    CMSEG(vector<array<I,0>> ig){}
    T get() {return v;}
    void update(T x){v=x;}
    T query() {return v;}
}; // eea5b0
```

## Strings (7)

**Hashing.h**  
Description: String hashing (multiple mods and  $2^{32}$ )  
Memory:  $\mathcal{O}(n)$   
Time:  $\mathcal{O}(1)$  query,  $\mathcal{O}(n)$  build

```
typedef uint64_t ull;
template<int M, class B> struct A {
    int x; B b; A(int x=0) : x(x), b(x) {}
    A(int x, B b) : x(x), b(b) {}
    A operator+(A o) const {int y = x+o.x; return{y - (y>=M)*M, b +o.b};}
    A operator-(A o) const {int y = x-o.x; return{y + (y<0)*M, b -o.b};}
    A operator*(A o) const { return {(int) ((ll)x*o.x % M), b*o.b} ; }
    explicit operator ull() const { return x ^ (ull) b << 21; }
}; // e03276
typedef A<1000000007, A<1000000009, unsigned>> H;
static int C = 9548; // initialize to a number less than MOD or random
struct HashInterval {
    int n; vector<H> ha, pw;
    template<typename S>
    HashInterval(const S & str) : n(sz(str)), ha(n+1), pw(n+1) {
        pw[0] = 1;
        rep(i,0,n)
            ha[i+1] = ha[i] * C + str[i],
            pw[i+1] = pw[i] * C;
    } // 185a86
    H query(int a, int b) { return ha[b+1] - ha[a] * pw[b-a+1]; }
    H queryI(int a, int b) { return query(n - b - 1, n - a - 1); }
}; // 1a81bd
```

**KMP.h**  
Description: KMP automaton  
Memory:  $\mathcal{O}(N)$   
Time:  $\mathcal{O}(N)$  build,  $\mathcal{O}(N+M)$  sequential querying

```
struct KMP {
    string P; int n; vector<int> nb;
    KMP(string& p) : P(p), n((int)P.size()), nb(n+1) {
        for(int k = 1; k < n; k++) nb[k+1] = nxt(nb[k], P[k]);
    } // ca6dc8
    int nxt(int i, char c){
        for(; i; i = nb[i])if (i < n and P[i]==c)return i+1;
        return P[0]==c;
    } // 2b99e2
    vector<vector<int>> dfa;
    void build_dfa(){ // uses a-z
        dfa.assign(n+1, vector<int>(26));
        dfa[0][P[0]-'a'] = 1;
        for(int k = 1; k <= n; k++)
            for(int c = 0; c < 26; c++)
                if (k < n and P[k] == 'a'+c) dfa[k][c] = k+1;
                else dfa[k][c] = dfa[nb[k]][c];
    }
```

```
    } // f47d83
}; // 8f8450

Zfunc.h
Description: z[i] computes the length of the longest common prefix of s[i] and s, except z[0] = 0. (abacaba -> 0010301)
Time:  $\mathcal{O}(n)$ 
ee09e2, 9L, 1m30s

vi Z(const string& S) {
    vi z(sz(S)); int l = -1, r = -1;
    rep(i,1,sz(S)) {
        z[i] = i >= r ? 0 : min(r - i, z[i - l]);
        while (i + z[i] < sz(S) && S[i + z[i]] == S[z[i]]) z[i]++;
        if (i + z[i] > r) l = i, r = i + z[i];
    } // 44be47
    return z;
} // ee09e2
```

**Manacher.h**  
Description: For each position in a string, computes  $p[0][i]$  = half length of longest even palindrome around pos i,  $p[1][i]$  = longest odd (half rounded down).  
Time:  $\mathcal{O}(N)$

```
array<vi, 2> manacher(const string& s) {
    int n = sz(s); array<vi,2> p = {vi(n+1), vi(n)};
    rep(z,0,2) for (int i=0,l=0,r=0; i < n; i++) {
        int t = r-i+!z;
        if (i<r) p[z][i] = min(t, p[z][l+t]);
        int L = i-p[z][i], R = i+p[z][i]-!z;
        while (L>=1 && R+1<n && s[L-1] == s[R+1]) p[z][i]++, L--, R++;
        if (R>r) l=L, r=R;
    } // a843d3
    return p;
} // e7ad79
```

**MinRotation.h**  
Description: Finds the lexicographically smallest rotation of a string.  
Usage: rotate(v.begin(), v.begin()+minRotation(v), v.end());  
Time:  $\mathcal{O}(N)$

```
int minRotation(string s) {
    int a=0, N=sz(s); s += s;
    rep(b,0,N) rep(k,0,N) {
        if (a+k == b || s[a+k] < s[b+k]) {b += max(0, k-1); break;}
        if (s[a+k] > s[b+k]) { a = b; break; }
    } // 9374b1
    return a;
} // d07a42
```

**Automaton.h**  
Description: Suffix automaton with parameterized transition table (can be array or map)  
Memory:  $\mathcal{O}(n*|G|)$   
Time:  $\mathcal{O}(n*accesstoG)$  build

```
template<class C, class G> struct Automaton{
    vector<G> dag;
    vector<int> lnk, len, cnt;
    int last;
    template<class S> Automaton(S& str):
        dag(2),lnk(2,0),len(2,0),cnt(2,0),last(1){
            for(C& c : str)add(c); //can do .reserve(2*n+2)
        } // cd5999
    int new_node(int l, int is_pref){
        dag.pb(G()); lnk.pb(0); len.pb(l); cnt.pb(is_pref);
        return sz(dag)-1;
    }
```

```

} // 0814f6
void add(C& c){
    int cur = new_node(len[last]+1, 1), p = last;
    while(p and not dag[p][c])dag[p][c] = cur, p = lnk[p];
    if (not p)lnk[cur] = 1;
    else {
        int q = dag[p][c];
        if (len[p]+1 == len[q])lnk[cur] = q;
        else{
            int clone = new_node(len[p]+1, 0);
            dag[clone] = dag[q];
            lnk[clone] = lnk[q];
            lnk[q] = lnk[cur] = clone;
            while(p and dag[p][c] == q)dag[p][c]=clone, p=lnk[p];
        } // 270a8a
    } // 9e6a19
    last = cur;
} // b1dc88
}; // 70eb49
```

**Aho.h**  
**Description:** Aho-Corasick  
**Memory:**  $\mathcal{O}(n * 26)$   
**Time:**  $\mathcal{O}(n)$  build

ccf3a0, 47L, 6m40s

```

const int K = 26;
struct Vertex {
    int next[K];
    bool output = false;
    int p = -1;
    char pch;
    int link = -1;
    int go[K];

    Vertex(int _p=-1, char ch='$') : p(_p), pch(ch) {
        fill(begin(next), end(next), -1);
        fill(begin(go), end(go), -1);
    } // 2636d9
}; // 0c589a
```

```
vector<Vertex> t(1);
```

```

void add_string(string const& s) {
    int v = 0;
    for (char ch : s) {
        int c = ch - 'a';
        if (t[v].next[c] == -1) {
            t[v].next[c] = sz(t);
            t.eb(v, ch);
        } // 2abd5c
        v = t[v].next[c];
    } // fc1e0d
    t[v].output = true;
} // 1bb4a6
```

```

int go(int v, char ch);
int get_link(int v) {
    if (t[v].link == -1) {
        if (v == 0 || t[v].p == 0) t[v].link = 0;
        else t[v].link = go(get_link(t[v].p), t[v].pch);
    } // e9856f
    return t[v].link;
} // 1e38ab
```

```

int go(int v, char ch) {
    int c = ch - 'a';
    if (t[v].go[c] == -1) {
        if (t[v].next[c] != -1) t[v].go[c] = t[v].next[c];
        else t[v].go[c] = v == 0 ? 0 : go(get_link(v), ch);
    }
```

```

} // 202af4
return t[v].go[c];
} // e442d8
```

DP (8)

We don't know DP optimizations yet :/

Various (9)

9.1 Random

**RNG.h**  
**Description:** RNGs  
**Time:**  $\mathcal{O}(1)$

6823af, 18L, 3m50s

```

mt19937 rng(chrono::steady_clock::now().time_since_epoch().
    count()); // mt19937-64
void example() {
    int n; cin >> n;
    uniform_int_distribution<int> distribution(1,n);
    int num = distribution(rng); // num no range [1, n]
    vector<int> vec(n);
    iota(vec.begin(), vec.end(), 1);
    shuffle(vec.begin(), vec.end(), rng); // shuffle
} // a3f379
```

```

using ull = unsigned long long;
ull mix(ull o){
    o+=0x9e3779b97f4a7c15;
    o=(o^(o>>30))*0xbf58476d1ce4e5b9;
    o=(o^(o>>27))*0x94d049bb133111eb;
    return o^(o>>31);
} // bc6211
ull hash(pii a) {return mix(a.first ^ mix(a.second));}
```

9.2 Ternary Search

**GoldenSectionSearch.h**  
**Description:** Finds the argument minimizing the function  $f$  in the interval  $[a,b]$  assuming  $f$  is unimodal on the interval, i.e. has only one local minimum and no local maximum. The maximum error in the result is  $\epsilon$ ps. Works equally well for maximization with a small change in the code. See Ternary-Search.h in for a discrete version.  
**Usage:** double func(double x) { return 4+x+.3\*x\*x; }  
double xmin = gss(-1000,1000,func);  
**Time:**  $\mathcal{O}(\log((b-a)/\epsilon))$

31d45b, 16L, 3m20s

```

// It is important for r to be precise,
// otherwise we don't necessarily maintain the inequality a <
    x1 < x2 < b.
double gss(double a, double b, double (*f)(double)) {
    double r = (sqrt(5)-1)/2, eps = 1e-7;
    double x1 = b - r*(b-a), x2 = a + r*(b-a);
    double f1 = f(x1), f2 = f(x2);
    while (b-a > eps)
        if (f1 < f2) { //change to > to find maximum
            b = x2; x2 = x1; f2 = f1;
            x1 = b - r*(b-a); f1 = f(x1);
        } else { // 4513d0
            a = x1; x1 = x2; f1 = f2;
            x2 = a + r*(b-a); f2 = f(x2);
        } // 2fe74a
    return a;
} // 31d45b
```

**TernarySearch.h**  
**Description:** Find the smallest  $i$  in  $[a,b]$  that maximizes  $f(i)$ , assuming that  $f(a) < \dots < f(i) \geq \dots \geq f(b)$ . To reverse which of the sides allows non-strict inequalities, change the  $<$  marked with (A) to  $\leq$ , and reverse the loop at (B). To minimize  $f$ , change it to  $>$ , also at (B).  
**Usage:** int ind = ternSearch(0,n-1,&(int i){return a[i];});  
**Time:**  $\mathcal{O}(\log(b-a))$

afb9, 9L, 1m30s

```

template<class F> int ternSearch(int a, int b, F f) {
    while (b - a >= 5) {
        int mid = (a + b) / 2;
        if (f(mid) < f(mid+1)) a = mid; // (A)
        else b = mid+1;
    } // ce7859
    rep(i,a+1,b+1) if (f(a) < f(i)) a = i; // (B)
    return a;
} // afbac9
```

9.3 Optimization tricks

9.3.1 Bit hacks

- $x \& -x$  is the least bit in  $x$ .
- for (int x = m; x; ) { --x &= m; ... } loops over all subset masks of  $m$  (except  $m$  itself).
- $c = x \& -x$ ,  $r = x + c$ ;  $((r \wedge x) \gg 2) / c$  |  $r$  is the next number after  $x$  with the same number of bits set.
- rep(b,0,K) rep(i,0,(1 << K))  
if (i & 1 << b) D[i] += D[i^(1 << b)];  
computes all sums of subsets.

9.3.2 Pragmas

- **#pragma** GCC optimize ("Ofast") will make GCC auto-vectorize loops and optimizes floating points better.
- **#pragma** GCC target ("avx2") can double performance of vectorized code, but causes crashes on old machines.
- **#pragma** GCC optimize ("bmi,bmi2,lzcnt,popcnt") are good for bitsets.